

PROJECT - EVENTFORCE MANAGEMENT SYSTEM

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EVENTFORCE MANAGEMENT SYSTEM- SALESFORCE IMPLEMENTATION

User Story:

The EventForce Management System is a Salesforce-based solution for managing Events, Clients, Vendors, Venues, and Feedback in a centralized platform. It automates key processes such as client reminders, event cancellation approvals, venue availability updates, and prevention of double bookings using Flows, Approval Processes, Apex Classes, and Triggers. The system also provides Reports, Dashboards, and role-based security to improve event coordination, data accuracy, operational visibility, and customer service.

1. INTRODUCTION

1.1 Business Overview

The EventForce Management System is a Salesforce-based solution developed to manage Events, Clients, Vendors, Venues, and Feedback in a centralized platform. The system uses custom objects, relationships, validation rules, Flows, Approval Processes, Apex Classes and Triggers to automate event management activities.

The solution supports automated client reminders, event cancellation approvals, venue availability updates, double-booking prevention, and automatic completion of past events. Reports, Dashboards, Profiles, Roles, Permission Sets, and Sharing Rules provide better operational visibility and controlled access to event data.

Overall, the project demonstrates how Salesforce can be customized to support event planning requirements through automation, security, data management, and reporting. It provides a centralized environment that helps reduce manual activities and improves the efficiency of day-to-day event management operations.

1.2 Purpose

The purpose of the EventForce Management System is to:

- Centralize Events, Clients, Vendors, Venues, and Feedback in Salesforce.
- Reduce manual work through automated Flows and Apex processes.
- Prevent duplicate venue bookings for the same date.
- Automate client reminders for confirmed events.
- Manage event cancellation requests through an Approval Process.
- Maintain accurate venue availability based on event status.
- Provide controlled access using Profiles, Roles, Permission Sets, and Sharing Rules.
- Improve operational visibility through Reports and Dashboards.

1.3 Objectives

Specific points:

- Centralize Event, Client, Vendor, Venue and Feedback data.
- Automate client reminders for confirmed events.
- Manage cancellation requests through an Approval Process.
- Prevent duplicate venue bookings.
- Maintain venue availability based on event status.
- Provide controlled access through Profiles, Roles, Permission Sets and Sharing Rules.
- Provide Reports and Dashboards for operational visibility.

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Event Operations Manager	Maintain all event-related information in one system	Event, client, vendor, and venue details can become	Information is maintained across different areas of	Concerned about data accuracy

			difficult to track	event operations	
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PS-2	Event Coordinator	Schedule events with the correct client and venue	Selecting and maintaining related records manually can cause errors	Event information depends on multiple related records	Worried about incorrect bookings
PS-3	Venue Manager	Worried about incorrect bookings	Venue status may not change automatically when an event is confirmed or canceled	Venue availability depends on event status	Concerned about availability
PS-4	Vendor Manager	Manage vendor information and coordinate vendors for events	Vendor details and event-vendor assignments need to be maintained separately	Different events may require different vendors and service types	Concerned about vendor coordination and assignment accuracy
PS-5	Event Coordinator	Notify clients before their confirmed events	Sending reminders manually can be missed or delayed	Upcoming events require timely communication	Stressed about missed reminders

PS-6	Event Administrator	Process event cancellation requests in a controlled manner	Cancellation decisions need approval before changing the event status	Different users may be involved in the cancellation process	Uncertain about cancellation decisions
PS-7	Event Coordinator	Maintain client feedback with the related event	Feedback can become difficult to connect with the correct client and event	Feedback depends on both client and event information	Concerned about customer experience
PS-8	Event Operations Manager	Monitor upcoming events and operational information	Important event information is difficult to summarize quickly	Event data needs to be presented through reports and dashboards	Less confident in operational decisions



2.2 Empathy Map Canvas

Persona: Event Coordinator

Role: Event Planning & Coordination

Focus: Connecting clients, events, venues, and vendors while keeping event activities on track.

Think & Feel:

Needs a clear view of every event and wants to complete coordination without scheduling errors or missed client communication.

Hear:

Gets event requirements, changes, cancellation requests, and service-related updates from clients, vendors, venue teams, and administrators.

See:

Works with client records, event schedules, venue details, vendor assignments, event status, feedback, reports, and dashboard information.

Say & Do:

Creates and updates event information, coordinates required vendors and venues, monitors event status, handles client communication, and follows up on upcoming events.

Pain Points:

Managing several related records, preventing venue conflicts, remembering client reminders, handling cancellation requests, and keeping operational information updated can require considerable manual effort.

Expected Gains:

A connected Salesforce system that automates reminders and approvals, maintains venue status, prevents duplicate venue bookings, and provides reports and dashboards for easier event monitoring.



2.3 Brainstorming & Idea Prioritization

Ideation – Solution Comparison

Idea	Approach	Key Limitation
Paper-Based Management	Maintain client, event, venue, and vendor information through physical records and registers.	Difficult to update, search, share, and maintain accurate information.
Spreadsheet Tracking	Store event schedules, client details, venue information, and vendor data in spreadsheets.	Limited relationship management, manual updates, and higher risk of duplicate or outdated data.
Custom Web Application	Build an independent application specifically for event booking, vendor coordination, venue management, and feedback.	Requires considerable development, testing, hosting, security, and ongoing maintenance.
Salesforce CRM	Build EventForce using custom objects, relationships, automation, security controls, reports, and dashboards	Requires proper Salesforce configuration, automation design, and user access setup.

Selected Approach – Salesforce Platform

Salesforce was chosen as the platform for implementing the EventForce Management System because it can bring the complete event-management process into one connected environment.

Key Reasons:

- **Process Automation:** Client reminders, cancellation approvals, venue status updates, and event completion can be automated.
- **Connected Data:** Events can be linked with Clients, Venues, Vendors, and Feedback through Salesforce relationships.
- **Booking Control:** Apex automation can help prevent multiple events from being assigned to the same venue on the same date.
- **Access Management:** Profiles, Roles, Permission Sets, Organization-Wide Defaults, and Sharing Rules provide controlled access to information.

- Operational Monitoring: Reports and dashboards provide a consolidated view of upcoming events and operational information.
- Future Expansion: The system can be extended with additional Salesforce automation, objects, reports, and business requirements.

Chosen Solution:

Salesforce Platform was selected as the foundation for EventForce because it allows event operations, automation, security, and reporting to be managed together within a single platform.

3. Requirement Analysis

3.1 Business Requirements

The EventForce Management System is designed to support the day-to-day activities involved in event planning and coordination. The system provides a centralized Salesforce environment for maintaining event, client, vendor, venue, and feedback information.

The major business requirements of the system are:

- Maintain event information including event date, event type, event status, client, venue, and budget.
- Maintain complete client information such as email, phone, address, country, and city.
- Maintain vendor details including service type, contact information, and availability status.
- Maintain venue information including address, location, capacity, and availability status.
- Connect events with the appropriate clients and venues.
- Associate vendors with events through the Event Vendor junction object.
- Maintain feedback related to the appropriate client and event.
- Send automated reminders to clients before confirmed events.
- Manage event cancellation requests through an approval process.
- Automatically update venue availability based on event status.
- Prevent two events from being assigned to the same venue on the same date.
- Automatically mark past events as Completed.
- Provide reports and dashboards for monitoring upcoming events and operational information.

- Control access to event information using Profiles, Roles, Permission Sets, Organization-Wide Defaults, and Sharing Rules.

3.2 Functional Requirements

The functional requirements define the major functions that the EventForce Management System should perform.

FR No	Requirement
FR-1	The system shall allow users to create and manage event records.
FR-2	The system shall maintain client information and associate clients with events.
FR-3	The system shall maintain vendor information including service type and status.
FR-4	The system shall maintain venue information including capacity and availability status.
FR-5	The system shall allow vendors to be associated with events using the Event Vendor junction object.
FR-6	The system shall maintain feedback records related to clients and events.
FR-7	The system shall validate client email addresses before saving invalid email information.
FR-8	The system shall calculate the event budget based on the selected event type.
FR-9	The system shall restrict the Event lookup in Feedback based on the selected client.
FR-10	The system shall send a reminder to the client three days before a confirmed event.

FR-11	The system shall process event cancellation requests through an approval process.
FR-12	The system shall update the event status to Canceled when a cancellation request is approved.

3.3 Non-Functional Requirements

NFR	Description
NFR-01	Usability: The system should provide a simple and organized interface for managing event-related information.
NFR-02	Security: Access to Salesforce objects and records should be controlled according to user roles and permissions.
NFR-03	Data Accuracy: Validation rules and automated processes should help maintain accurate event and client information.
NFR-04	Reliability: Automated processes should perform the required actions consistently when their conditions are met.
NFR-05	Maintainability: Flows, Apex classes, triggers, approval processes, and configurations should be structured so they can be monitored and maintained.
NFR-06	Scalability: The solution should support additional events, clients, vendors, venues, users, and future business requirements.
NFR-07	Performance: Automated processing should reduce unnecessary manual activities and improve operational efficiency.
NFR-08	Reporting: The system should provide clear reports and dashboard information for operational monitoring.

3.4 Data Model

The EventForce Management System uses a connected Salesforce data model to organize the information required for event planning and coordination.

The system contains the following custom objects:

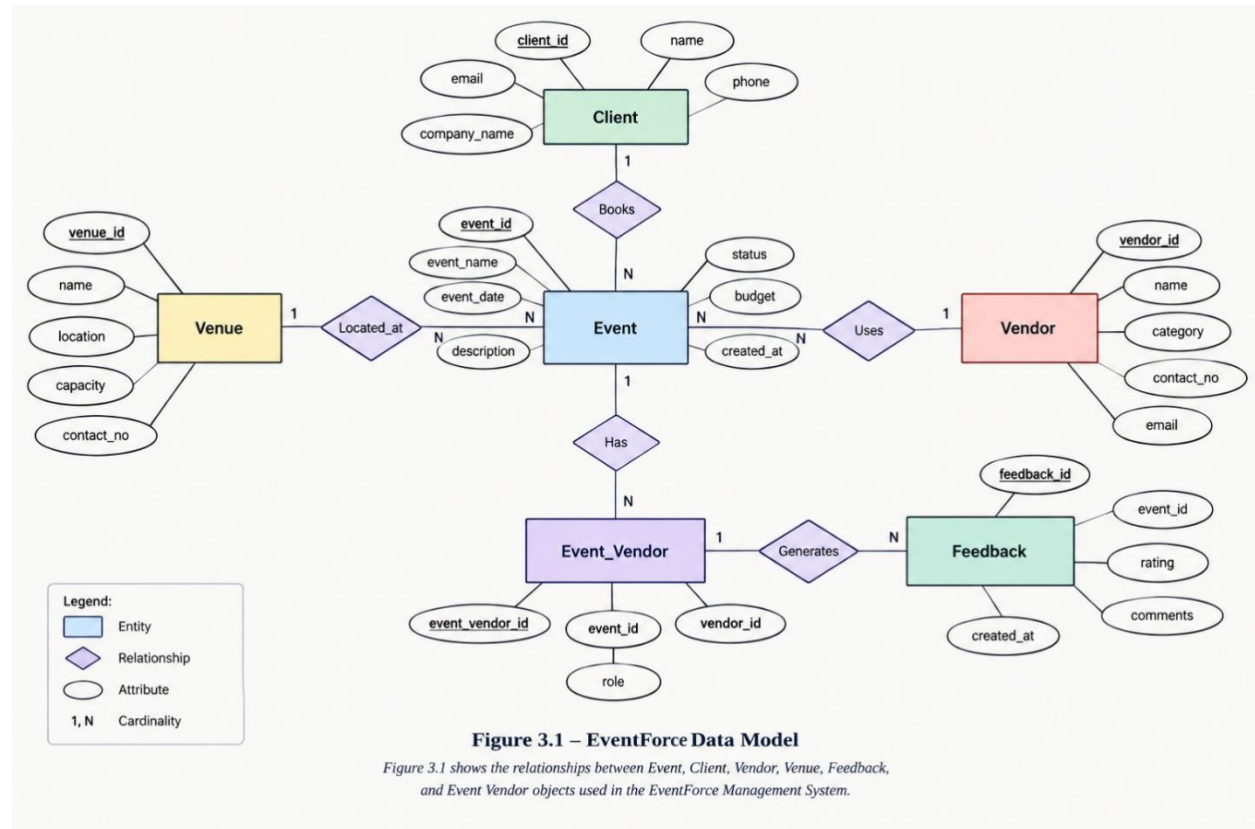
- Event – Stores event details such as Event Name, Event Date, Event Type, Event Status, Client, Venue, and Event Budget.
- Client – Stores client contact and location information.
- Vendor – Stores vendor contact information, service type, and status.
- Venue – Stores venue address, location, capacity, and availability status.
- Feedback – Stores client feedback and rating information related to an event.
- Event Vendor – Acts as a junction object to associate Events with Vendors.

Object Relationships

- Event → Client: Lookup relationship used to associate an event with a client.
- Event → Venue: Lookup relationship used to associate an event with a venue.
- Feedback → Event: Lookup relationship used to connect feedback with an event.
- Feedback → Client: Lookup relationship used to connect feedback with a client.
- Event Vendor → Event: Master-Detail relationship connecting the junction record to an event.
- Event Vendor → Vendor: Master-Detail relationship connecting the junction record to a vendor.

3.5 ER Diagram

The relationships between the EventForce objects are represented in the Entity Relationship Diagram.

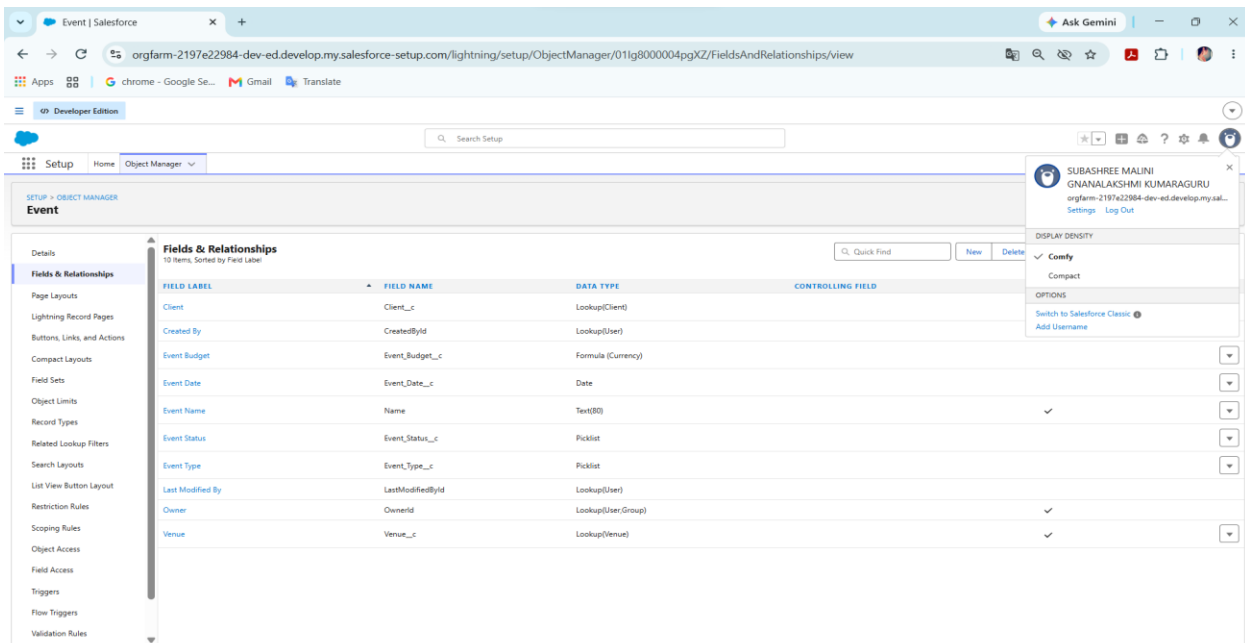


4. PHASE 1 – REQUIREMENT ANALYSIS & PLANNING

4.1 Custom Objects

4.1.1 Event Object

The **Event** object is used to store and manage the main event information in the EventForce Management System. It contains details such as Event Name, Event Date, Event Type, Event Status, Event Budget, Client, and Venue. The Event object acts as the central object for managing event scheduling and related event activities.

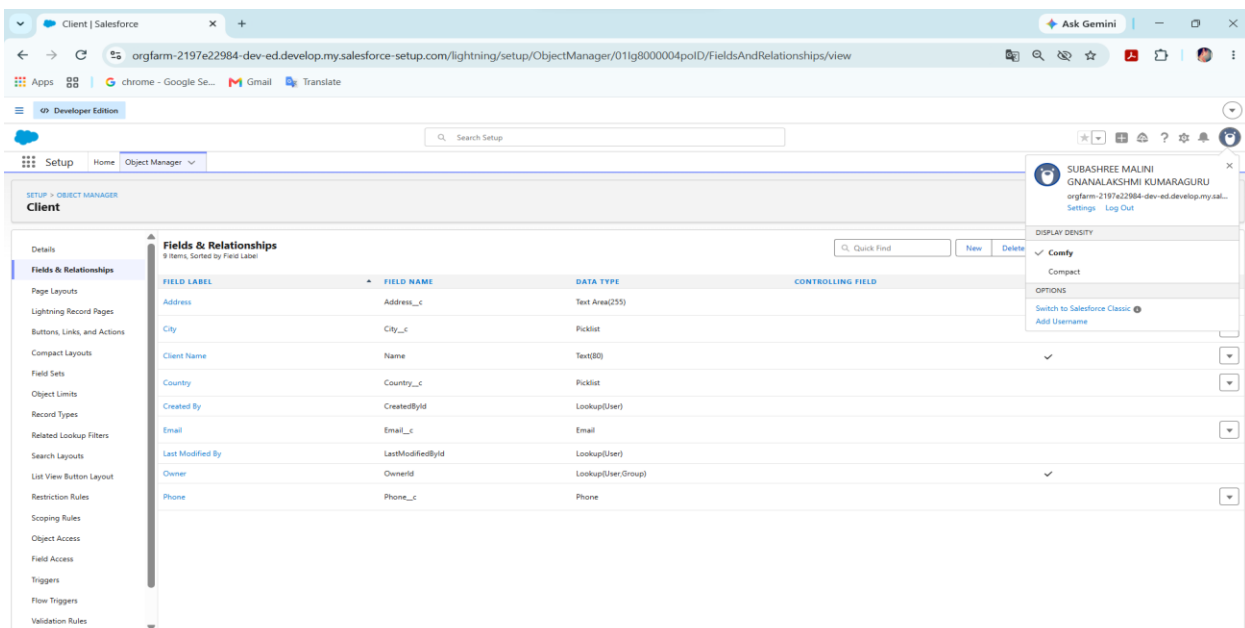


The screenshot shows the Salesforce Object Manager interface for the 'Event' object. The left sidebar contains a navigation menu with options like Details, Fields & Relationships, Page Layouts, and others. The main area displays the 'Fields & Relationships' section for the 'Event' object, showing a table of fields with their labels, names, data types, and whether they are controlling fields. A user profile dropdown is visible in the top right corner.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD
Client	Client__c	Lookup(Client)	
Created By	CreatedById	Lookup(User)	
Event Budget	Event_Budget__c	Formula (Currency)	
Event Date	Event_Date__c	Date	
Event Name	Name	Text(80)	✓
Event Status	Event_Status__c	Picklist	
Event Type	Event_Type__c	Picklist	
Last Modified By	LastModifiedById	Lookup(User)	
Owner	OwnerId	Lookup(User/Group)	✓
Venue	Venue__c	Lookup(Venue)	✓

4.1.2 Client Object

The **Client** object is used to maintain customer information related to events. It stores details such as Client Name, Email, Phone, Address, Country, and City. The Client object is linked with the Event object so that each event can be associated with the corresponding client.

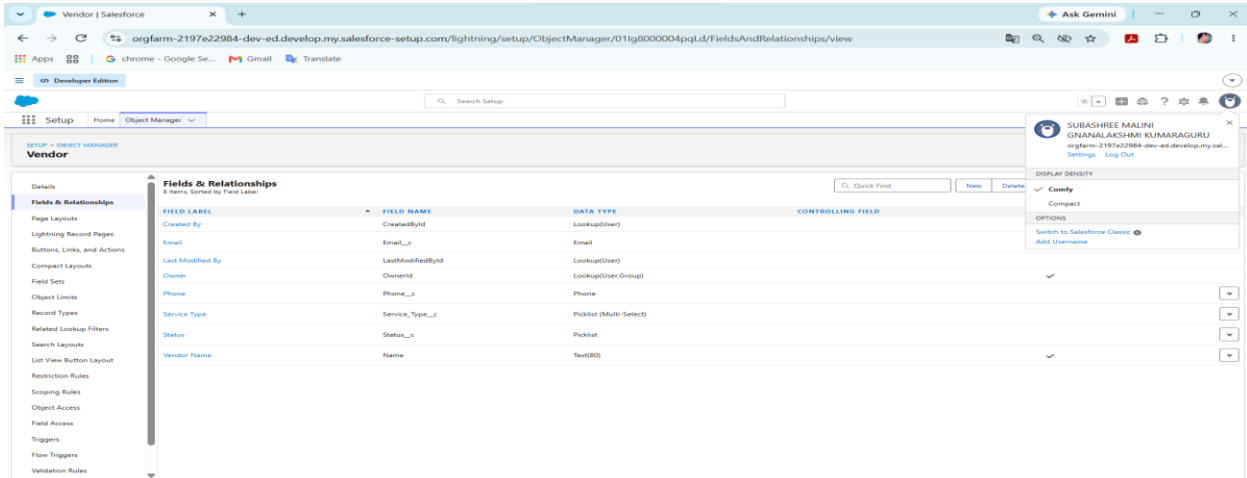


The screenshot shows the Salesforce Object Manager interface for the 'Client' object. The left sidebar contains a navigation menu with options like Details, Fields & Relationships, Page Layouts, and others. The main area displays the 'Fields & Relationships' section for the 'Client' object, showing a table of fields with their labels, names, data types, and whether they are controlling fields. A user profile dropdown is visible in the top right corner.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD
Address	Address__c	Text Area(255)	
City	City__c	Picklist	
Client Name	Name	Text(80)	✓
Country	Country__c	Picklist	
Created By	CreatedById	Lookup(User)	
Email	Email__c	Email	
Last Modified By	LastModifiedById	Lookup(User)	
Owner	OwnerId	Lookup(User/Group)	✓
Phone	Phone__c	Phone	

4.1.3 Vendor Object

The **Vendor** object is used to maintain information about vendors who provide services for events. It contains Vendor Name, Email, Phone, Service Type, and Status. Vendor records can be associated with events through the Event Vendor junction object.

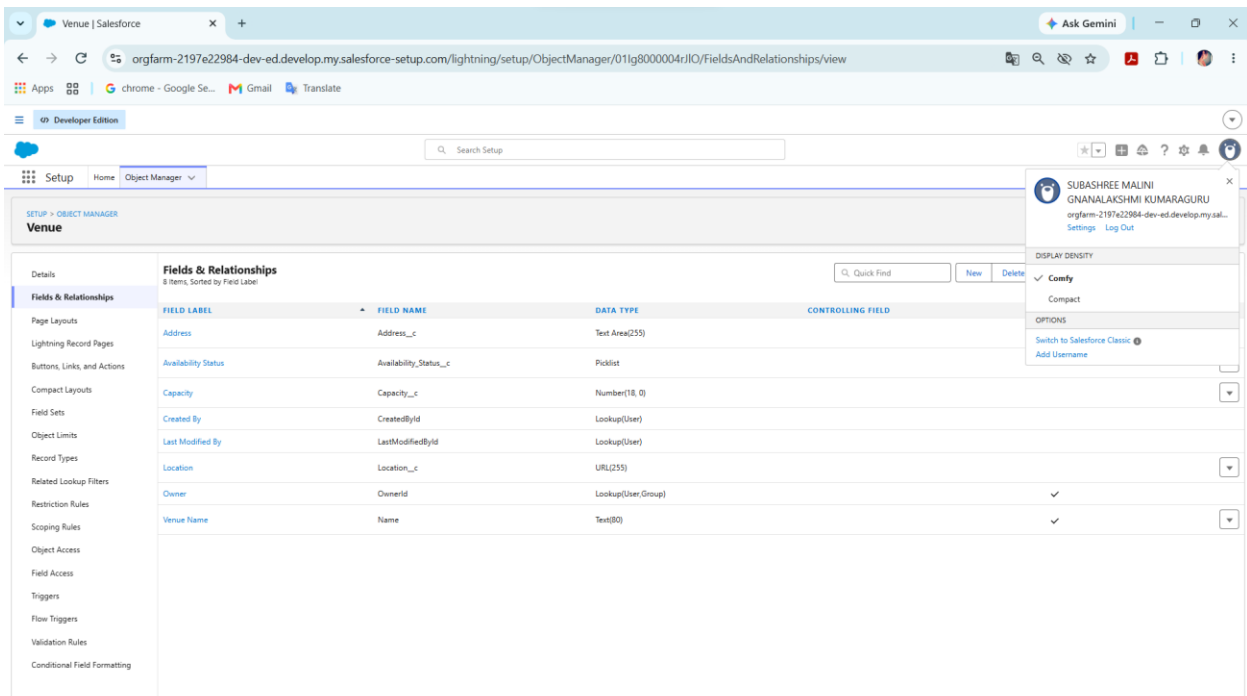


The screenshot shows the Salesforce Setup interface for the Vendor object. The left sidebar contains a navigation menu with options like Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, List View Button Layout, Restriction Rules, Scoping Rules, Object Access, Field Access, Triggers, Flow Triggers, and Validation Rules. The main content area displays the 'Fields & Relationships' section for the Vendor object, showing a table of fields with their labels, names, data types, and whether they are controlling fields.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD
Created By	CreatedById	Lookup(User)	
Email	Email_c	Email	
Last Modified By	LastModifiedById	Lookup(User)	
Owner	OwnerId	Lookup(User, Group)	
Phone	Phone_c	Phone	
Service Type	Service_Type__c	Picklist (Multi-Select)	
Status	Status__c	Picklist	
Vendor Name	Name	Text(80)	

4.1.4 Venue Object

The **Venue** object is used to store information about event venues. It contains Venue Name, Address, Location URL, Capacity, and Availability Status. The availability status is also updated by the EventForce automation when an event is confirmed or canceled.

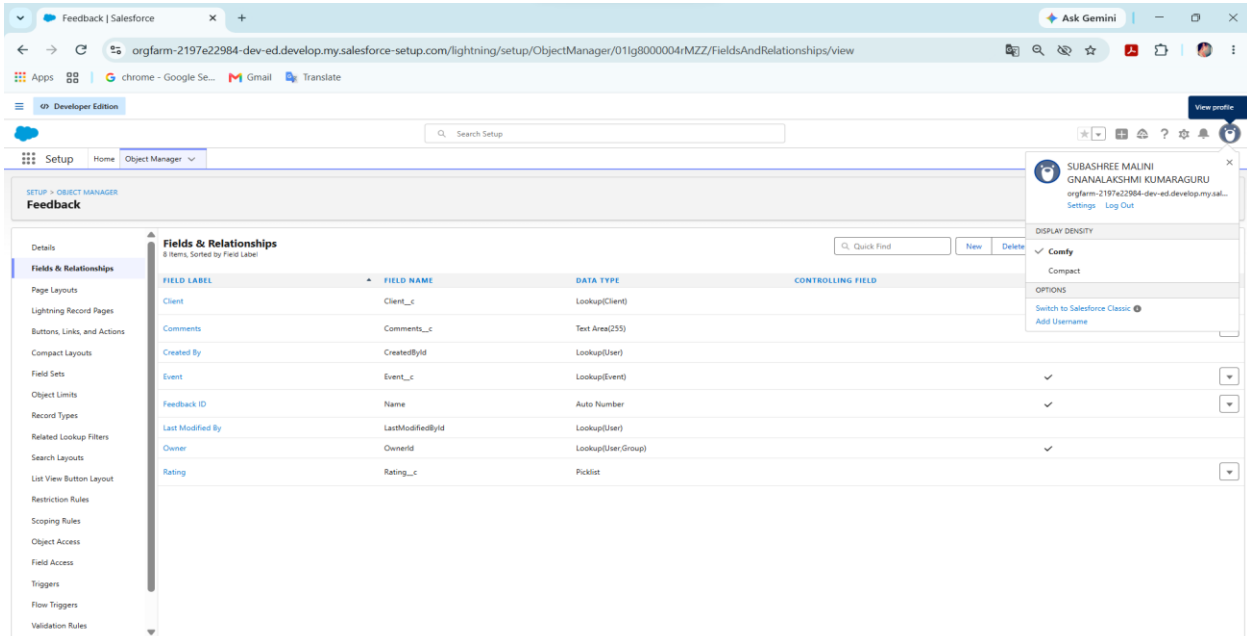


The screenshot shows the Salesforce Setup interface for the Venue object. The left sidebar contains a navigation menu with options like Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, List View Button Layout, Restriction Rules, Scoping Rules, Object Access, Field Access, Triggers, Flow Triggers, and Validation Rules. The main content area displays the 'Fields & Relationships' section for the Venue object, showing a table of fields with their labels, names, data types, and whether they are controlling fields.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD
Address	Address_c	Text Area(255)	
Availability Status	Availability_Status__c	Picklist	
Capacity	Capacity_c	Number(18, 0)	
Created By	CreatedById	Lookup(User)	
Last Modified By	LastModifiedById	Lookup(User)	
Location	Location_c	URL(255)	
Owner	OwnerId	Lookup(User, Group)	
Venue Name	Name	Text(80)	

4.1.5 Feedback Object

The **Feedback** object is used to capture feedback provided for completed or managed events. It contains Feedback ID, Rating, Comments, Client, and Event information. The Event and Client relationships help connect feedback with the appropriate customer and event.

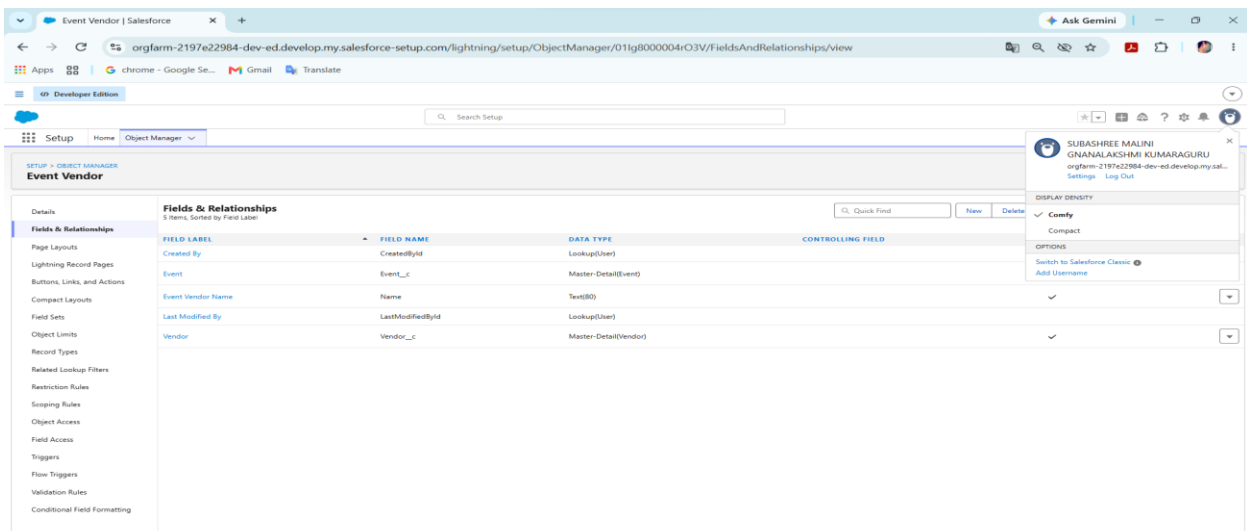


The screenshot shows the Salesforce Setup interface for the Feedback object. The left sidebar contains a navigation menu with options like Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, List View Button Layout, Restriction Rules, Scoping Rules, Object Access, Field Access, Triggers, Flow Triggers, and Validation Rules. The main content area is titled 'Feedback' and shows the 'Fields & Relationships' section. It displays a table with 8 items, sorted by Field Label. The table has columns for FIELD LABEL, FIELD NAME, DATA TYPE, and CONTROLLING FIELD. The fields listed are Client, Comments, Created By, Event, Feedback ID, Last Modified By, Owner, and Rating. A user profile dropdown is visible on the right, showing the user's name, email, and org name.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD
Client	Client_c	Lookup(Client)	
Comments	Comments_c	Text Area(255)	
Created By	CreatedById	Lookup(User)	
Event	Event_c	Lookup(Event)	
Feedback ID	Name	Auto Number	
Last Modified By	LastModifiedById	Lookup(User)	
Owner	OwnerId	Lookup(User/Group)	
Rating	Rating_c	Picklist	

4.1.6 Event Vendor Object

The **Event Vendor** object is a **junction object** used to establish a many-to-many relationship between Events and Vendors. An event can require multiple vendors, and a vendor can provide services for multiple events. The Event Vendor object uses **Master-Detail relationships** with both the Event and Vendor objects. This allows vendor assignments to be maintained for individual events.

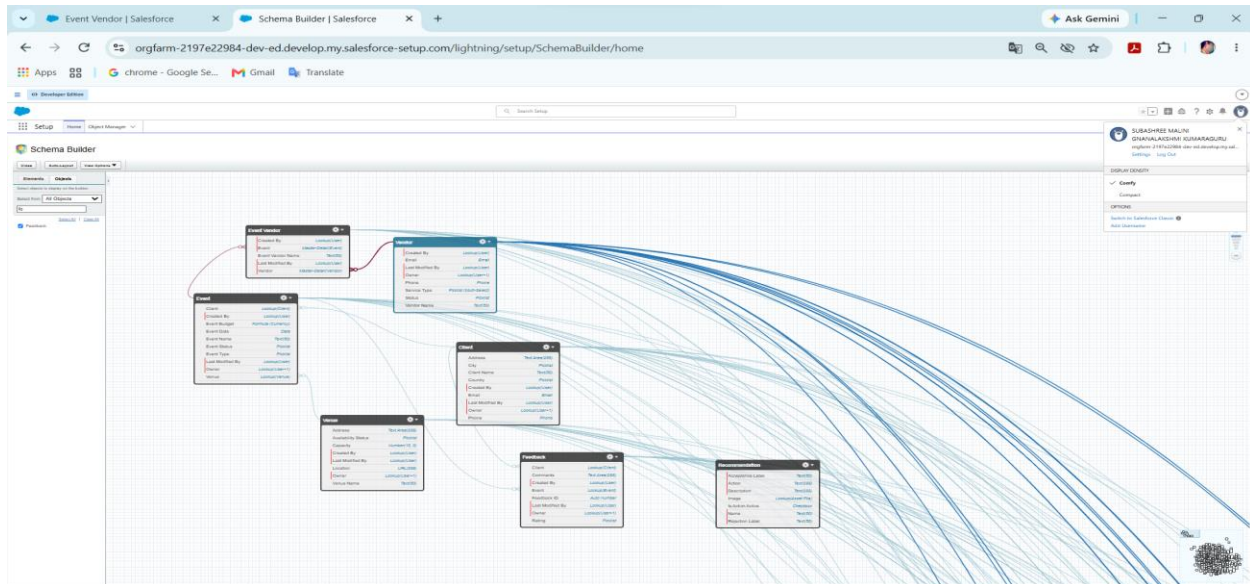


The screenshot shows the Salesforce Setup interface for the Event Vendor object. The left sidebar contains a navigation menu with options like Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Restriction Rules, Scoping Rules, Object Access, Field Access, Triggers, Flow Triggers, and Validation Rules. The main content area is titled 'Event Vendor' and shows the 'Fields & Relationships' section. It displays a table with 5 items, sorted by Field Label. The table has columns for FIELD LABEL, FIELD NAME, DATA TYPE, and CONTROLLING FIELD. The fields listed are Created By, Event, Event Vendor Name, Last Modified By, and Vendor. A user profile dropdown is visible on the right, showing the user's name, email, and org name.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD
Created By	CreatedById	Lookup(User)	
Event	Event_c	Master-Detail(Event)	
Event Vendor Name	Name	Text(80)	
Last Modified By	LastModifiedById	Lookup(User)	
Vendor	Vendor_c	Master-Detail(Vendor)	

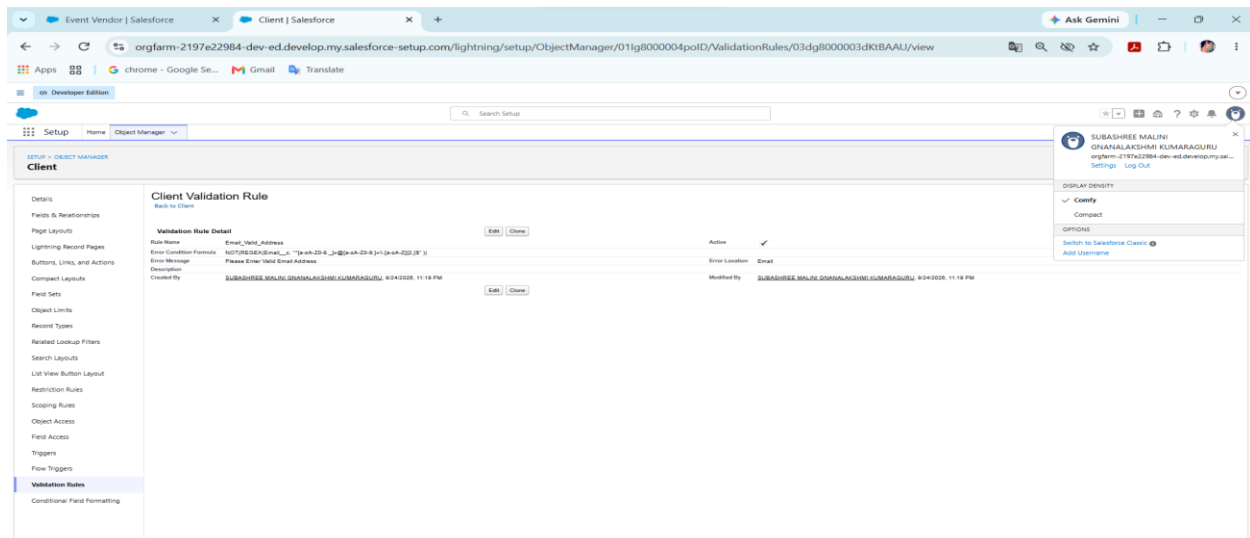
4.2 Relationships

The EventForce Management System uses Lookup and Master-Detail relationships to connect related records across the application. The Event object is linked to the Client and Venue objects through Lookup relationships. The Event Vendor junction object uses Master-Detail relationships with Event and Vendor to support the many-to-many relationship between events and vendors.



4.3 Validation Rule

The **Email_Valid_Address** validation rule is configured on the Client object to ensure that users enter a valid email address format. The rule checks the Email field using a regular expression and prevents the record from being saved when the entered email does not match the required format.



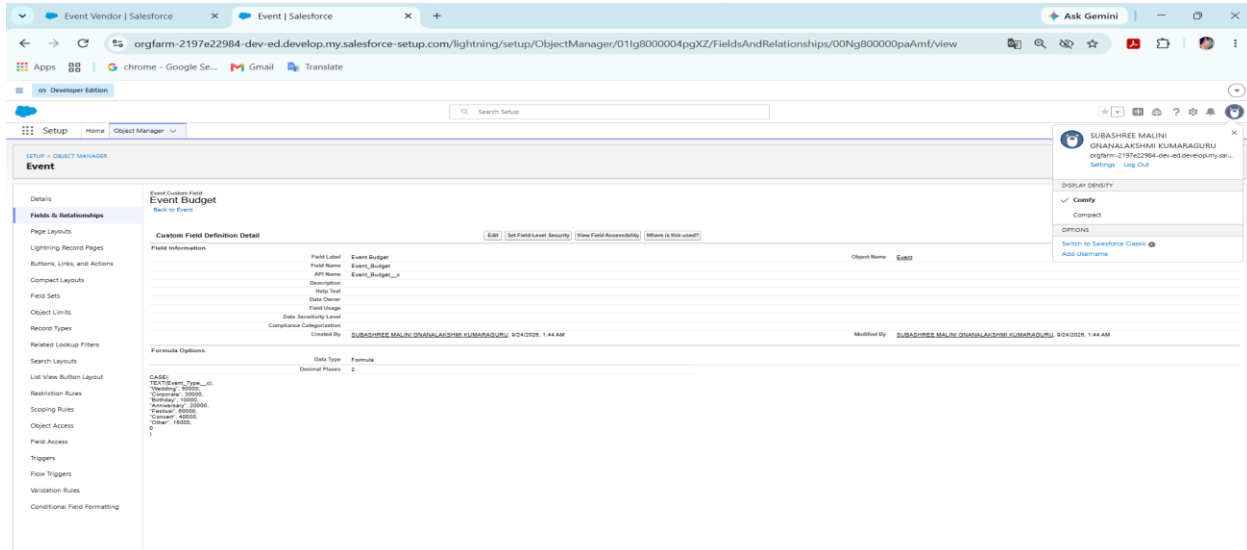
The screenshot shows the Salesforce 'Client Validation Rule' configuration page. The page is titled 'Client Validation Rule' and includes a 'Details' section on the left. The main content area displays the 'Validation Rule Detail' for the 'Email_Valid_Address' rule. The rule is configured with the following details:

- Rule Name:** Email_Valid_Address
- Error Condition Formula:** NOT(REGEX(Email, "[a-zA-Z0-9]+@[a-zA-Z0-9]+(\.[a-zA-Z0-9]+)"))
- Error Message:** Please Enter Valid Email Address
- Created By:** SUBASHREE MALINI GUNANAKRISHN KUNARAGURU
- Created On:** 8/24/2023, 11:18 PM
- Modified By:** SUBASHREE MALINI GUNANAKRISHN KUNARAGURU
- Modified On:** 8/24/2023, 11:18 PM

The page also includes a 'Details' section on the left with links to 'Fields & Relationships', 'Page Layouts', 'Lightning Record Pages', 'Buttons, Links, and Actions', 'Compact Layouts', 'Field Sets', 'Object Limits', 'Record Types', 'Related Lookup Filters', 'Search Layouts', 'List View Button Layout', 'Restriction Rules', 'Scoping Rules', 'Object Access', 'Field Access', 'Triggers', 'Flow Triggers', and 'Validation Rules'.

4.4 Formula Field

The **Event Budget** formula field is configured on the Event object to automatically determine the event budget based on the selected Event Type. Different budget values are assigned for Wedding, Corporate, Birthday, Anniversary, Festival, Concert, and Other events. This reduces manual budget entry and keeps the budget calculation consistent across event records.

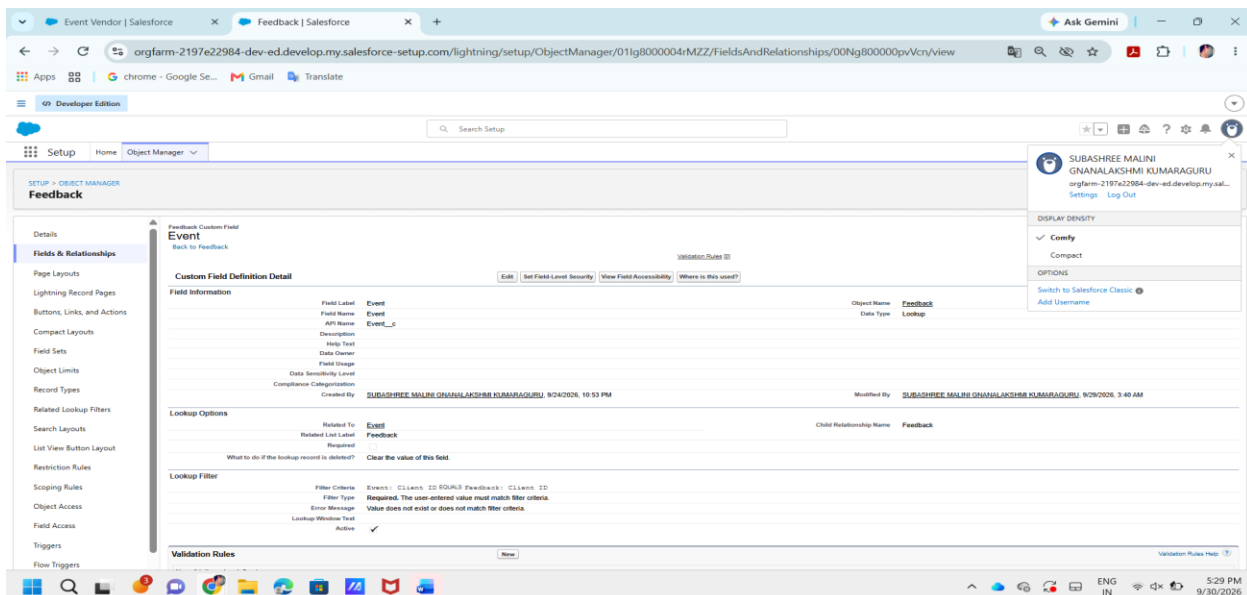


The screenshot shows the Salesforce Setup interface for the 'Event' object. The 'Fields & Relationships' section is selected, and the 'Custom Field Definition Detail' for the 'Event Budget' formula field is displayed. The field is of type 'Formula' and is calculated based on the 'Event Type' field. The formula options are defined as follows:

Event Type	Formula
Wedding	20000
Corporate	10000
Birthday	5000
Anniversary	25000
Festival	15000
Concert	10000
Other	10000

4.5 Lookup Filter

A lookup filter is configured on the **Event** lookup field of the Feedback object to display only the events associated with the selected client. The filter compares the Client field of the Event with the Client field selected on the Feedback record. This helps users select the appropriate event while entering feedback.



The screenshot shows the Salesforce Setup interface for the 'Feedback' object. The 'Fields & Relationships' section is selected, and the 'Custom Field Definition Detail' for the 'Event' lookup field is displayed. The field is of type 'Lookup' and is related to the 'Event' object. The lookup filter is configured to show only events where the 'Client' field matches the 'Client' field of the feedback record. The filter criteria are defined as follows:

Filter Criteria	Filter Type
Event: Client IS EQUAL Feedback: Client IS	Required. The user-entered value must match filter criteria.

The error message for the filter is: 'Value does not exist or does not match filter criteria.'

Project Tracker, Velocity & Burndown Chart:

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Event Management	US-01	As an Event Coordinator, I want to create events with type, date, and status so that event information is standardized.	5	High	Member1
Sprint-1	Client & Venue Management	US-02	As an Event Coordinator, I want to associate a client and venue with an event so that coordination is centralized.	5	High	Member1
Sprint-2	Vendor Management	US-03	As a Vendor Manager, I want to assign multiple vendors to an event so that services can be coordinated.	5	High	Member1
Sprint-2	Notifications & Automation	US-04	As a Client, I want timely event reminders so that I know when my event is approaching.	5	High	Member4
Sprint-3	Approval Management	US-05	As an Event Coordinator, I want cancellation requests to follow an approval process so that cancellations are controlled.	5	High	Member1
Sprint-3	Venue Management & Apex	US-06	As a Venue Manager, I want double bookings prevented so that venue schedules remain accurate.	8	High	Member2
Sprint-4	Security & Access Control	US-07	As an Event Administrator, I want role-based access so that users see only appropriate information.	5	High	Member3
Sprint-4	Reports & Dashboards	US-08	As Management, I want reports and dashboards so that event operations can be monitored.	5	Medium	Member1

5. PHASE 2 – BACKEND DEVELOPMENT & CONFIGURATIONS

5.1 Flows

Client Reminder – 3 Days Before

A record-triggered Flow was created to automatically send a reminder email to the client three days before a confirmed event. The Flow is triggered when an Event is created or updated with the Event Status set to **Confirmed**. A scheduled path named **3_Days_Before** uses the Event Date as the time source and is configured to run three days before the event. The Flow sends the reminder to the Client Email and copies the Event Owner.

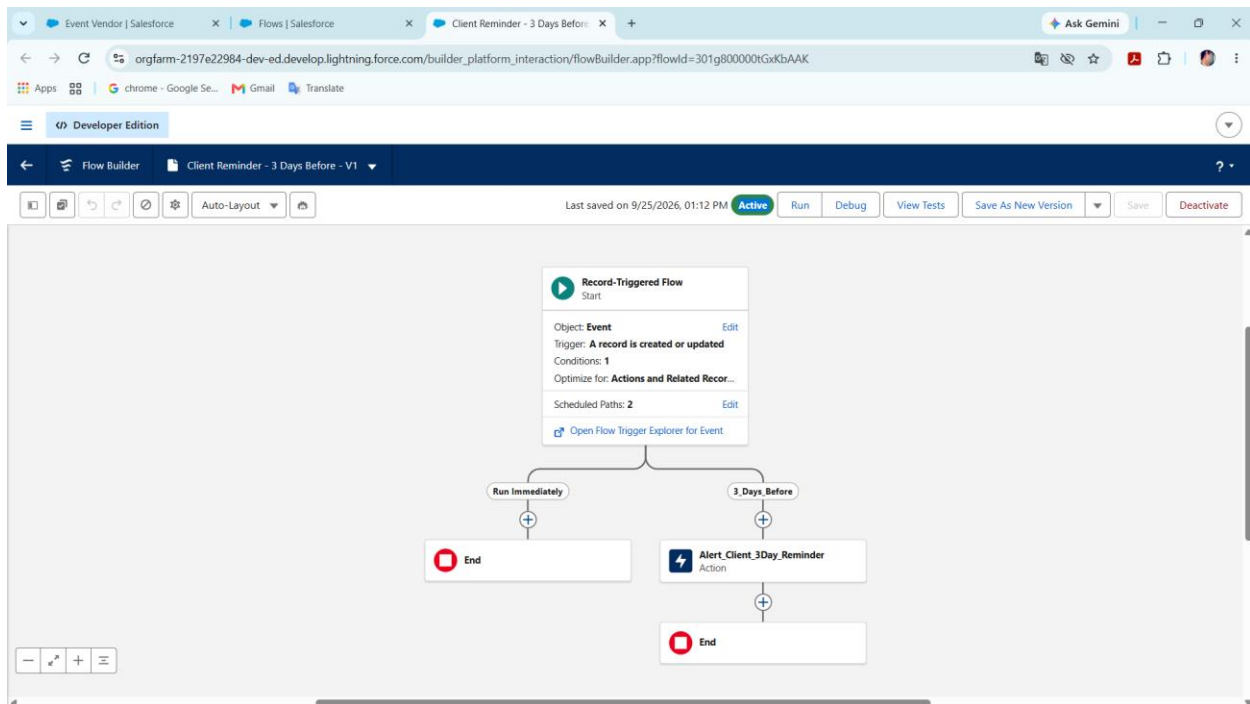


Figure 5.1 – Client Reminder – 3 Days Before Flow
&
Activated Client Reminder Flow

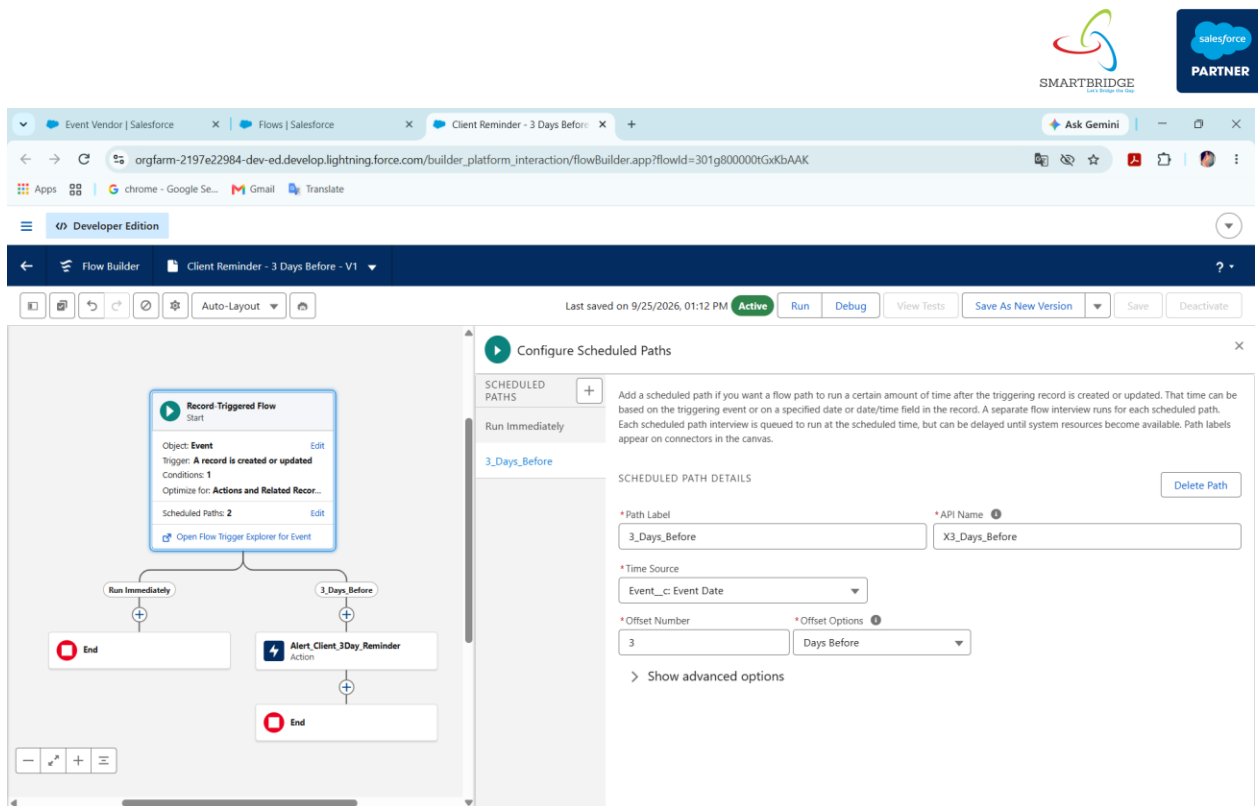


Figure 5.2 – Scheduled Path for Three-Day Client Reminder

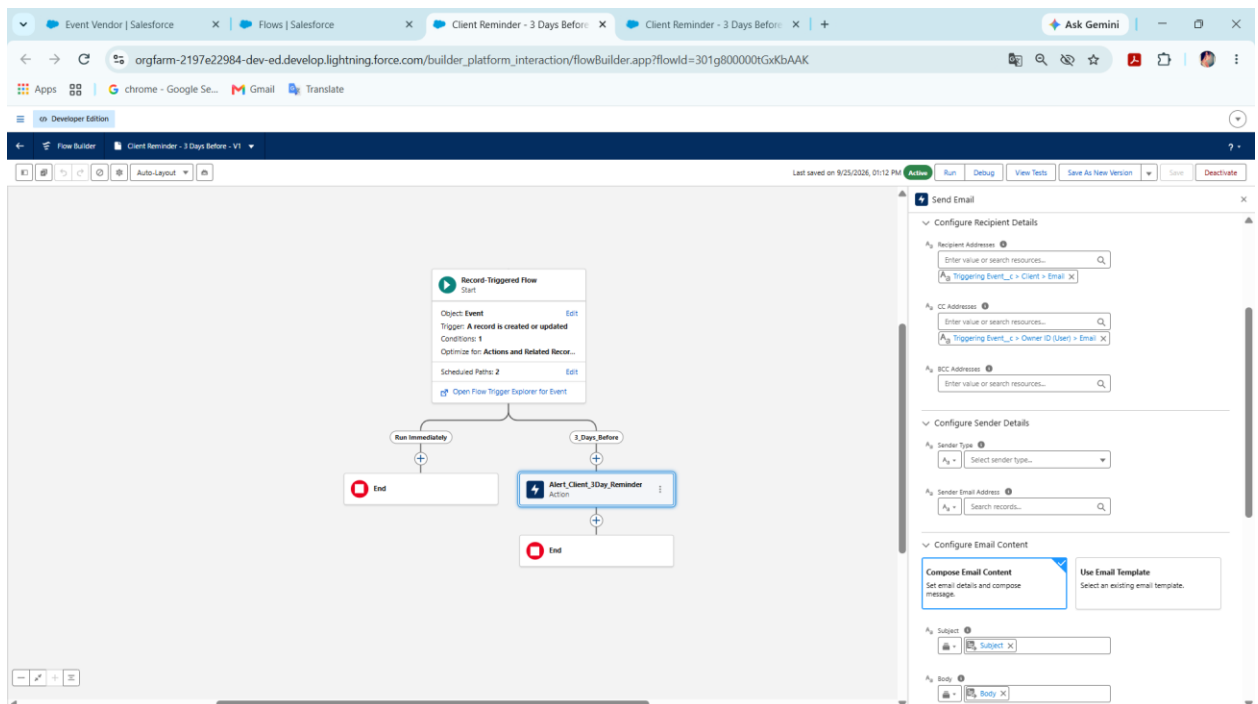
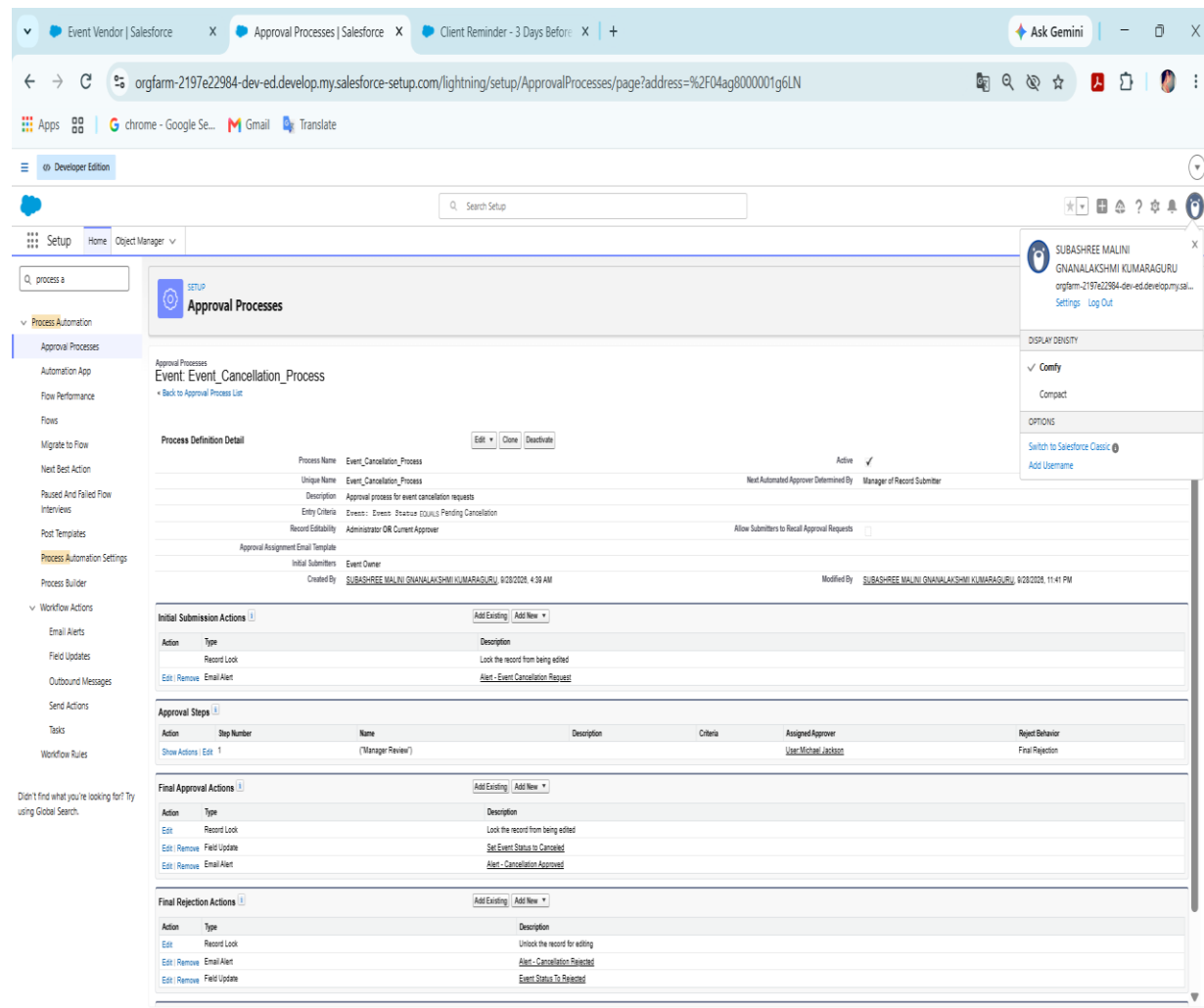


Figure 5.3 – Email Notification Configuration

5.2 Approval Process

Event Cancellation Approval Process

An Approval Process named **Event_Cancellation_Process** was configured for the Event object to manage event cancellation requests. The process is initiated when the Event Status is **Pending Cancellation**. The Event Owner submits the cancellation request for approval. The approval step is assigned to the designated approver. When approved, the Event Status is updated to **Canceled** and an approval notification is sent. When rejected, the Event Status is updated to **Rejected** and a rejection notification is sent.



The screenshot shows the Salesforce Setup interface for configuring an Approval Process. The left sidebar lists navigation options under 'Process Automation' and 'Workflow Actions'. The main content area displays the 'Event_Cancellation_Process' configuration.

Process Definition Detail

Field	Value
Process Name	Event_Cancellation_Process
Unique Name	Event_Cancellation_Process
Description	Approval process for event cancellation requests
Entry Criteria	Event: Event Status EQUALS Pending Cancellation
Record Editability	Administrator OR Current Approver
Approval Assignment Email Template	
Initial Submitters	Event Owner
Created By	SUBASHREE MALINI GNANALAKSHMI KUMARAGURU
Created On	9/20/2020 4:39 AM
Modified By	SUBASHREE MALINI GNANALAKSHMI KUMARAGURU
Modified On	9/20/2020 11:41 PM

Initial Submission Actions

Action	Type	Description
Record Lock	Record Lock	Lock the record from being edited
Email Alert	Email Alert	Alert - Event Cancellation Request

Approval Steps

Action	Step Number	Name	Description	Criteria	Assigned Approver	Reject Behavior
Show Actions	1	(Manager Review)			User: Michael Jackson	Final Rejection

Final Approval Actions

Action	Type	Description
Record Lock	Record Lock	Lock the record from being edited
Field Update	Field Update	Set Event Status to Canceled
Email Alert	Email Alert	Alert - Cancellation Approved

Final Rejection Actions

Action	Type	Description
Record Lock	Record Lock	Unlock the record for editing
Email Alert	Email Alert	Alert - Cancellation Rejected
Field Update	Field Update	Event Status To Rejected

Figure 5.4 – Event Cancellation Approval Process Configuration

5.3 Apex Class & Triggers

Venue Status and Double-Booking Prevention

Apex automation was configured to manage venue availability and prevent double booking. The VenueStatusHelper class updates the Venue Availability Status based on the Event Status. When an Event is Confirmed, the related Venue is marked as Reserved, and when an Event is Canceled, the Venue is marked as Available.

The EventTrigger13 trigger invokes the helper after Event insert or update. A separate PreventDoubleBooking trigger runs before Event insert or update and prevents booking the same Venue on the same date by displaying the message **“This Venue is already booked on this date.”**

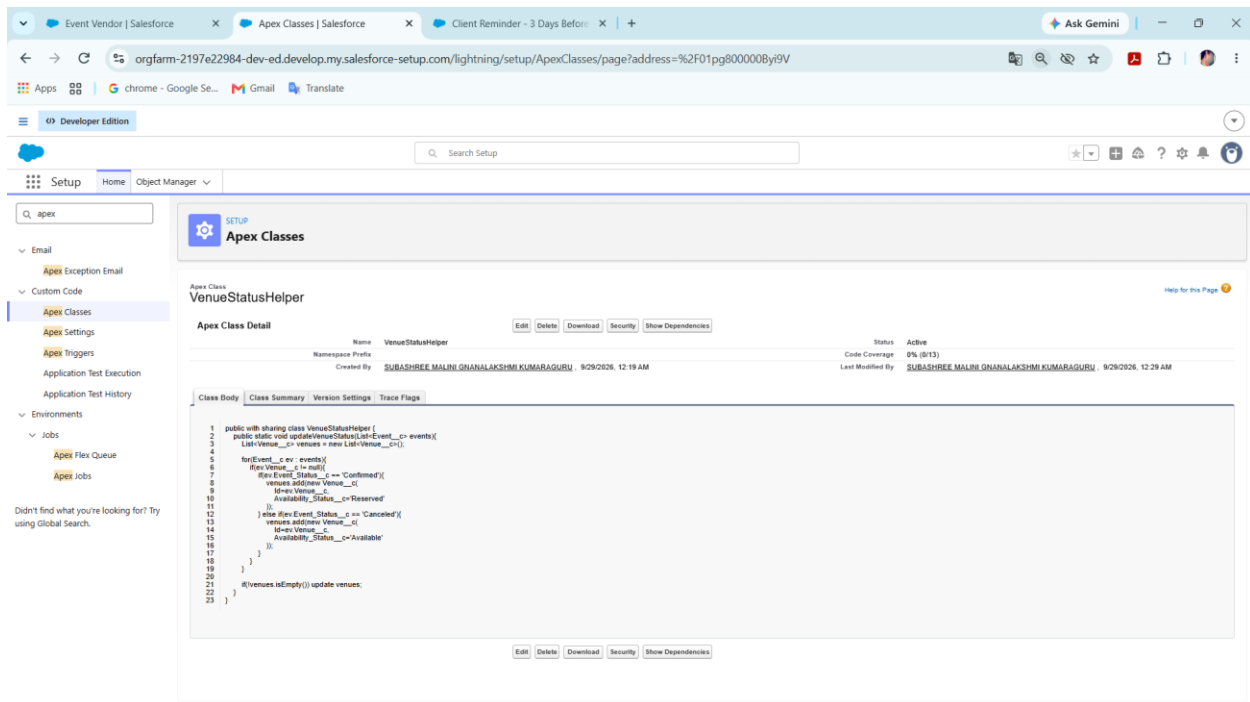


Figure 5.9 – Venue Status Helper Apex Class

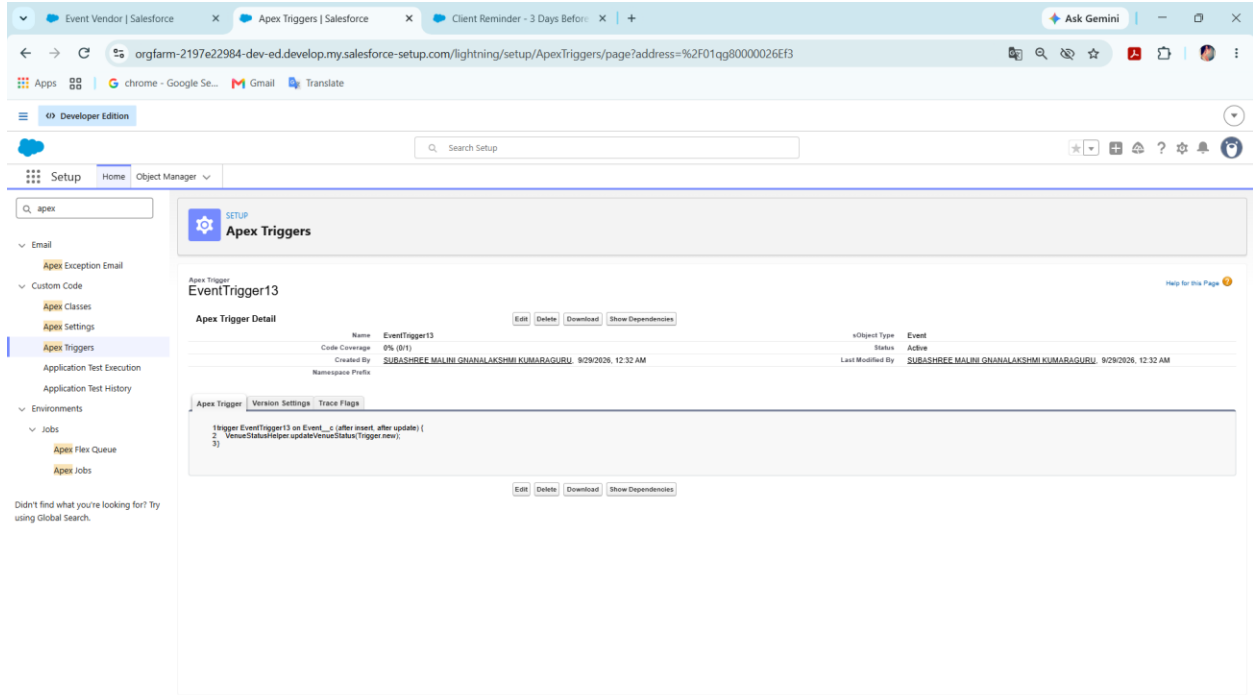


Figure 5.10 – Event Trigger for Venue Status Management

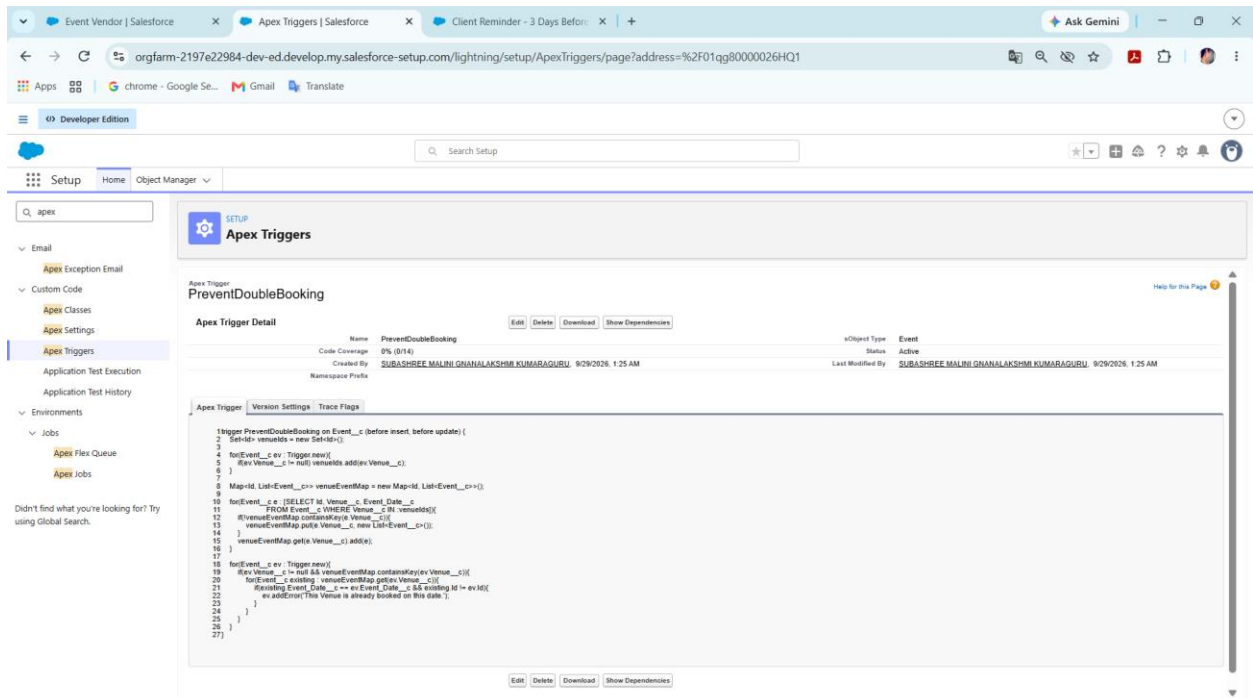
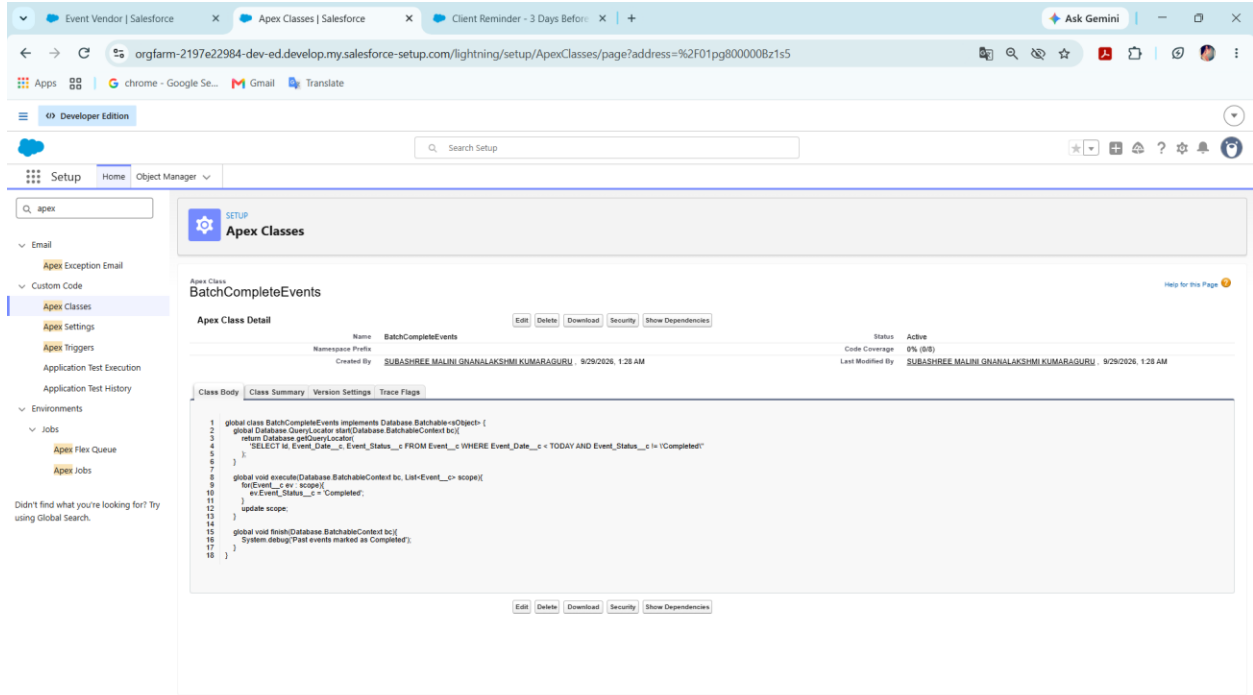


Figure 5.11 – Prevent Double Booking Trigger



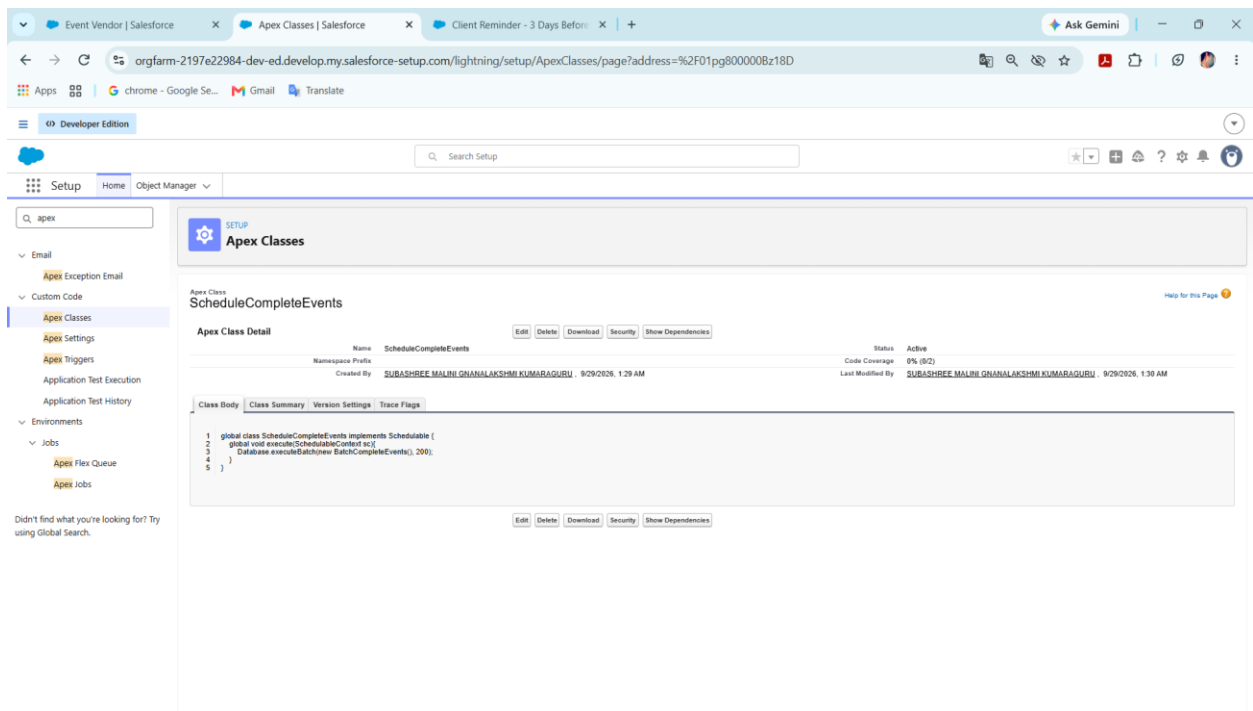
The screenshot shows the Salesforce Developer Edition interface. The left sidebar contains navigation links for Setup, Home, and Object Manager. The main content area displays the details for the Apex Class 'BatchCompleteEvents'. The class is created by SUBASHREE MALINI GNANALAKSHMI KUMARAGURU on 9/29/2026 at 1:28 AM. The class body contains the following code:

```

1 global class BatchCompleteEvents implements Database.Batchable<Object> {
2     global Database.QueryLocator start(Database.BatchableContext bc) {
3         return Database.query('SELECT id, Event_Date__c, Event_Status__c FROM Event__c WHERE Event_Date__c < TODAY AND Event_Status__c != \'Completed\'');
4     }
5 }
6
7 global void execute(Database.BatchableContext bc, List<Event__c> scope) {
8     for(Event__c ev : scope) {
9         ev.Event_Status__c = 'Completed';
10        update scope;
11    }
12 }
13
14 global void finish(Database.BatchableContext bc) {
15     System.debug('Past events marked as Completed');
16 }
17 }

```

Figure 5.12 – Batch Apex for Completing Past Events\



The screenshot shows the Salesforce Developer Edition interface. The left sidebar contains navigation links for Setup, Home, and Object Manager. The main content area displays the details for the Apex Class 'ScheduleCompleteEvents'. The class is created by SUBASHREE MALINI GNANALAKSHMI KUMARAGURU on 9/29/2026 at 1:29 AM. The class body contains the following code:

```

1 global class ScheduleCompleteEvents implements Schedulable {
2     global void execute(SchedulableContext sc) {
3         Database.executeBatchAsync(BatchCompleteEvents(), 200);
4     }
5 }

```

Figure 5.13 – Scheduled Apex for Automatic Event Completion

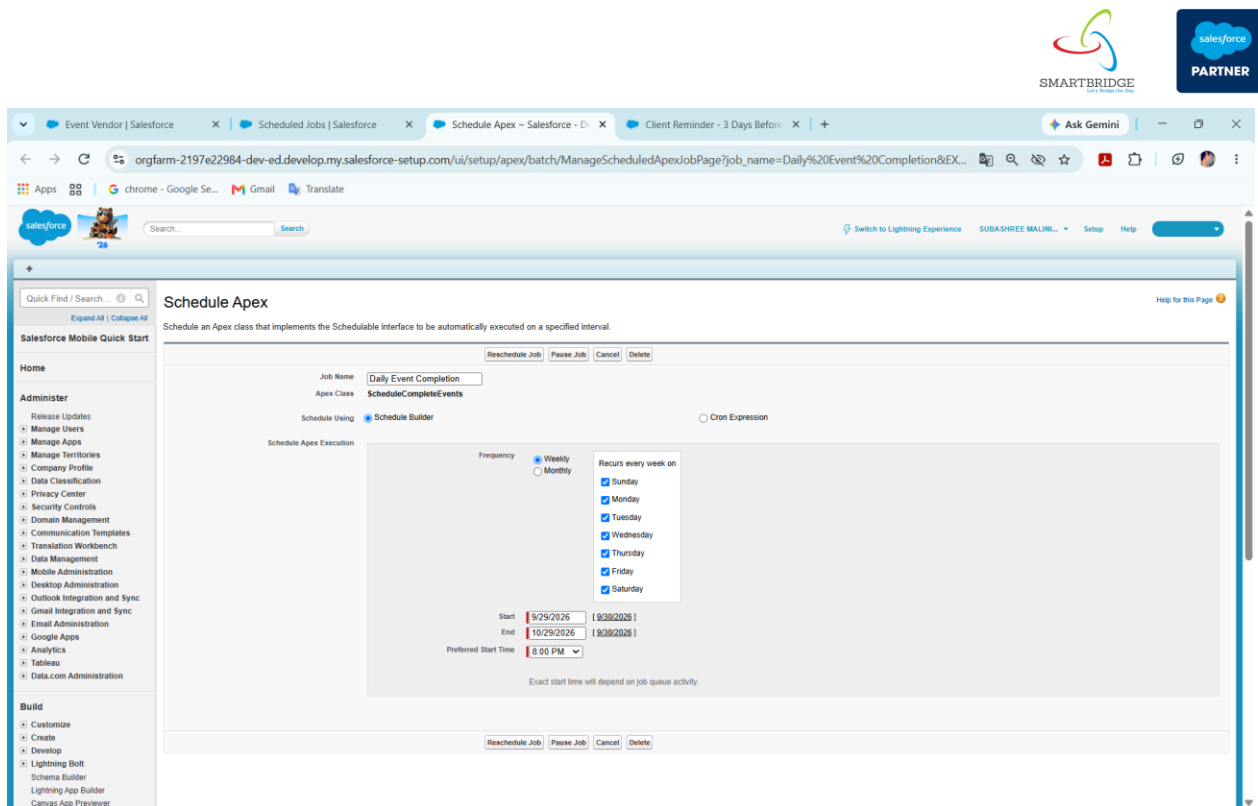


Figure 5.14 – Scheduled Apex Job for Daily Event Completion

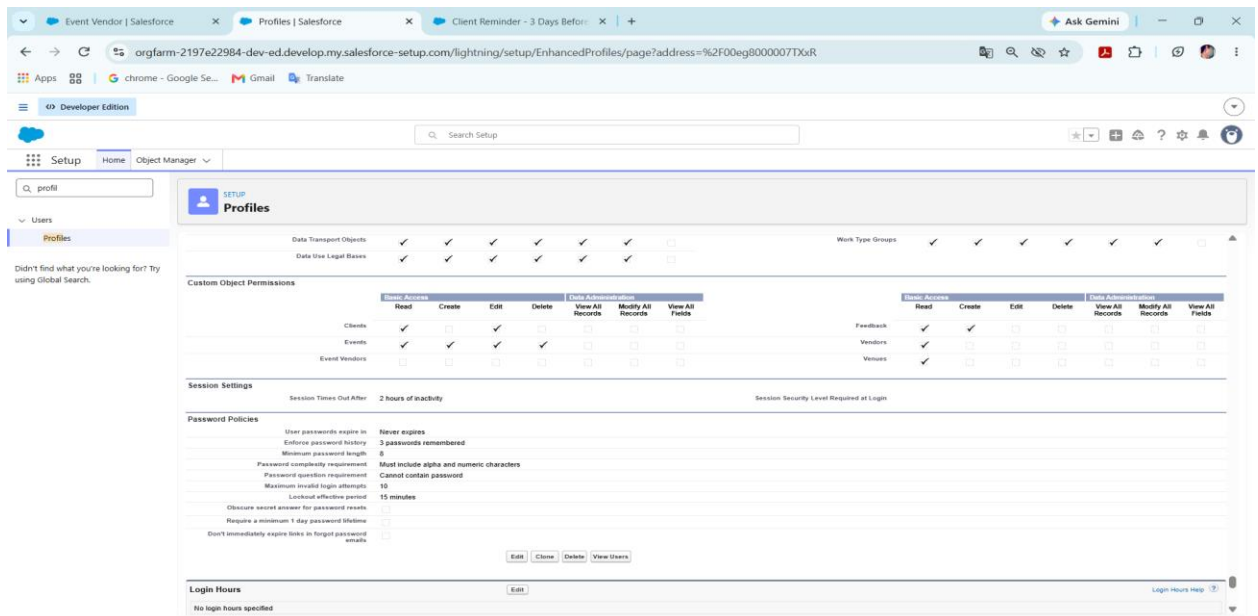


Figure 5.15 – Event Coordinator Profile and Object Permissions

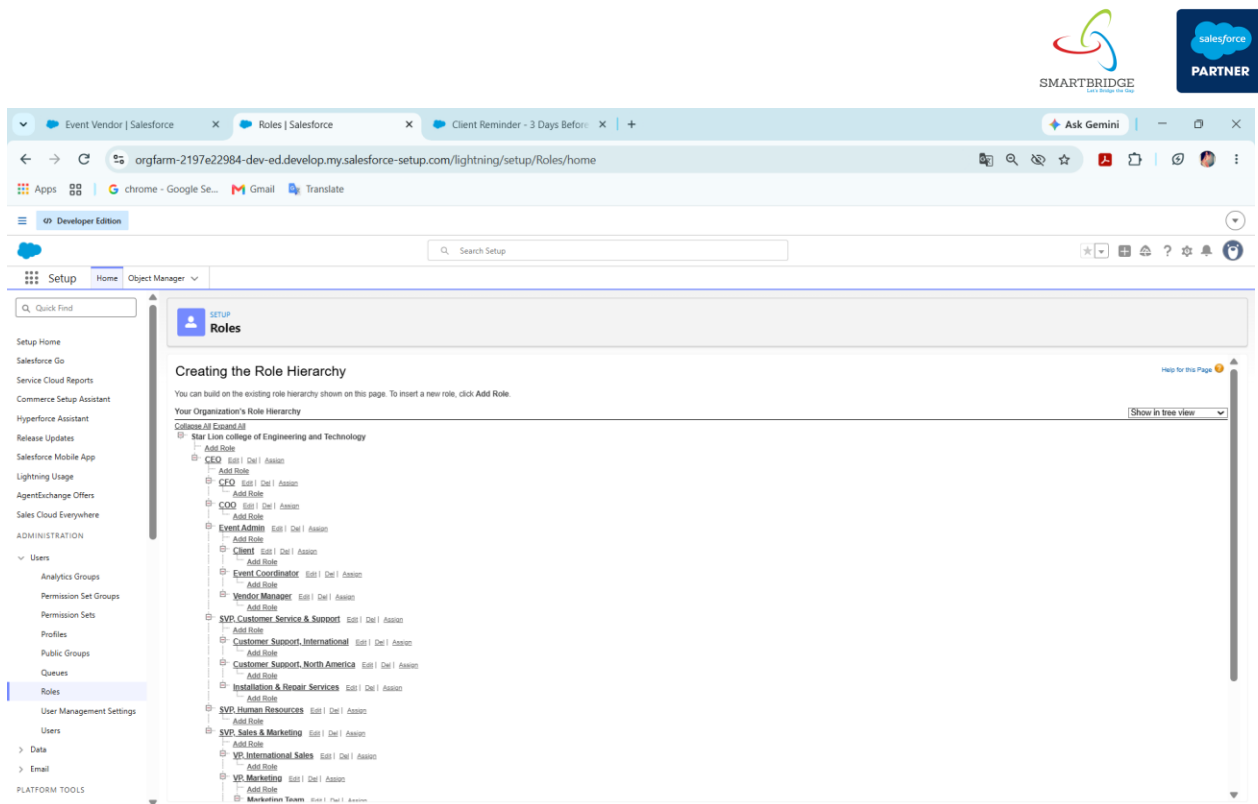


Figure 5.16 – EventForce Role Hierarchy

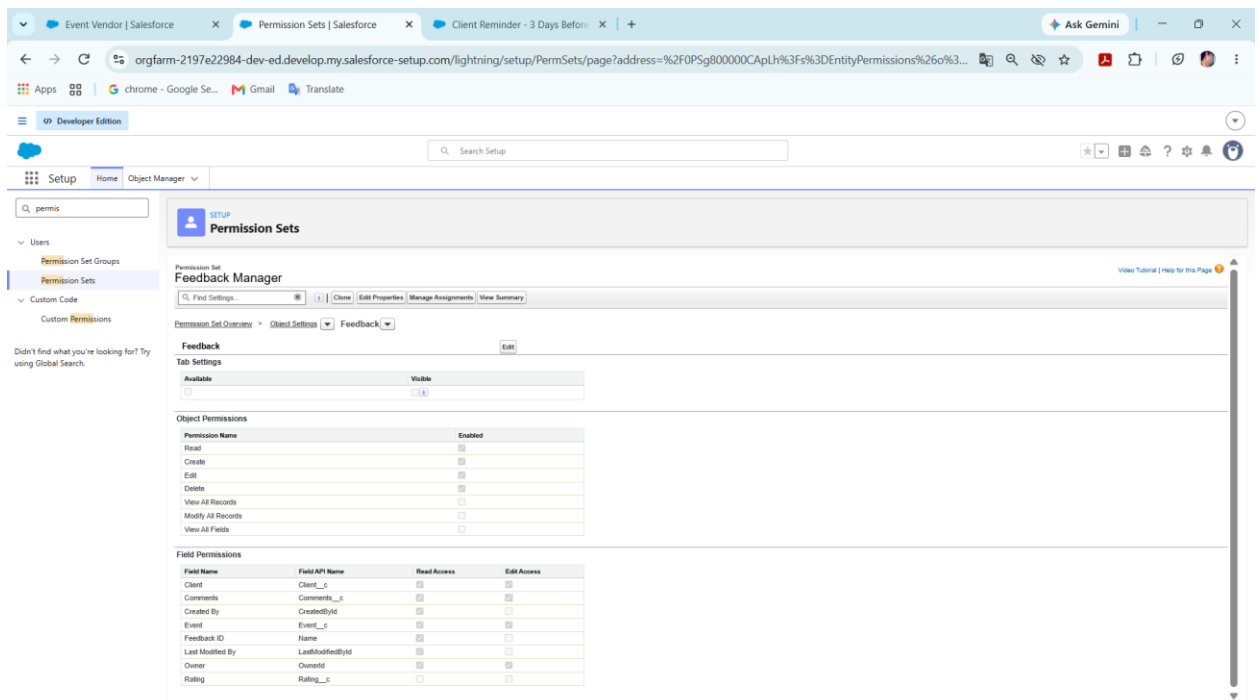


Figure 5.17 – Feedback Manager Permission Set

Sharing Settings

Object	Sharing Model	Access Level	Controlled by Parent
User Presence	Public Read Only	Private	✓
User Provisioning Request	Private	Private	✓
Utilization Alert	Public Read/Write	Private	✓
Wallist	Private	Private	✓
Web Card Document	Private	Private	✓
Work Item	Private	Private	✓
Work Item Test Suite Group Execution	Private	Private	✓
Work Order	Private	Private	✓
Work Plan	Private	Private	✓
Work Plan Template	Private	Private	✓
Work Step Template	Private	Private	✓
Work Type	Private	Private	✓
Work Type Group	Public Read/Write	Private	✓
Client	Public Read/Write	Private	✓
Event	Private	Private	✓
Event Vendor	Controlled by Parent	Controlled by Parent	✓
Feedback	Public Read/Write	Private	✓
Vendor	Public Read/Write	Private	✓
Venue	Public Read/Write	Private	✓

Other Settings

- Manager Groups: ☐ (1)
- Secure guest user record access: ☒ (1)
- Require permission to view record names in lookup fields: ☐ (1)
- Test asynchronous sharing recalculation in Apex tests: ☐ (1)
- Grant access using hierarchies by default in new queries: ☐ (1)

Sharing Rules

Event Sharing Rules

Sharing Settings

Event Sharing Rules

Rule Name	Criteria	Access Level
Event Sharing Rules	Owner in Role: Event Coordinator	Read Only

Other Sharing Rules

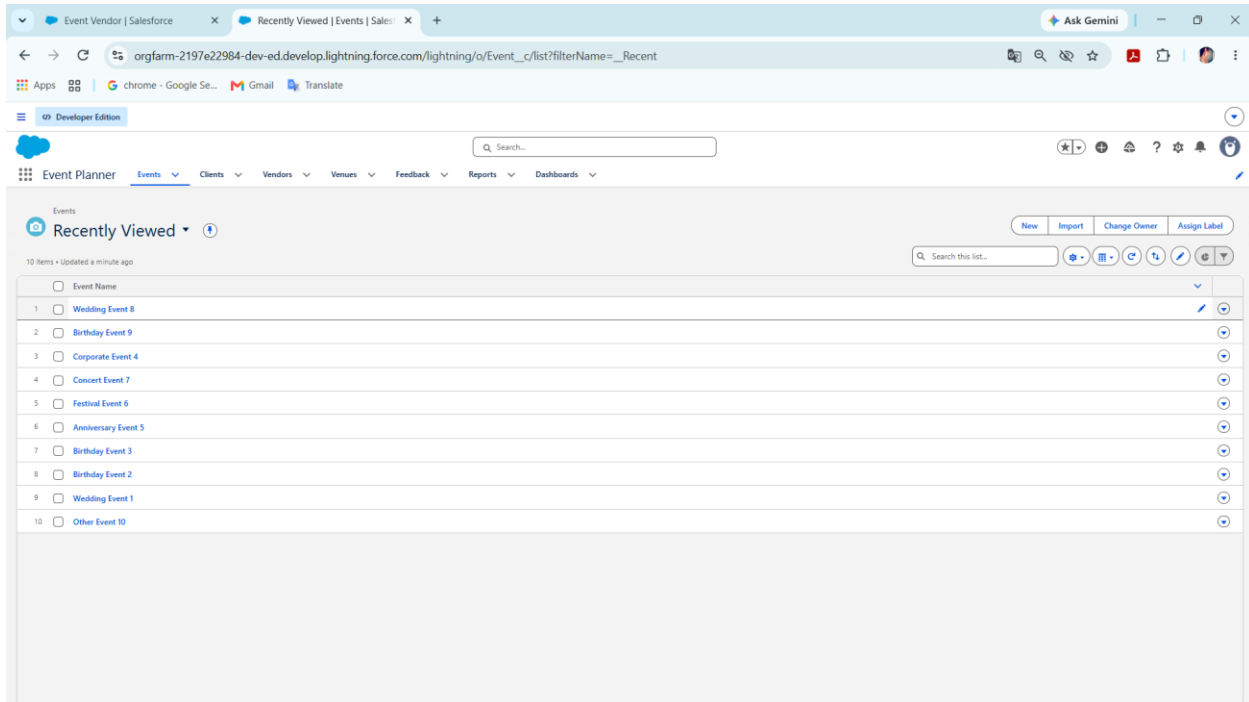
- Work Plan Template Sharing Rules: No sharing rules specified.
- Work Step Template Sharing Rules: No sharing rules specified.
- Work Type Sharing Rules: No sharing rules specified.
- Work Type Group Sharing Rules: No sharing rules specified.
- Client Sharing Rules: No sharing rules specified.
- Feedback Sharing Rules: No sharing rules specified.
- Vendor Sharing Rules: No sharing rules specified.
- Venue Sharing Rules: No sharing rules specified.

Figure 5.18 – Event Security and Sharing Configuration

6. PHASE 3 – UI/UX DEVELOPMENT & CUSTOMIZATION

6.1 Lightning App – Event Planner

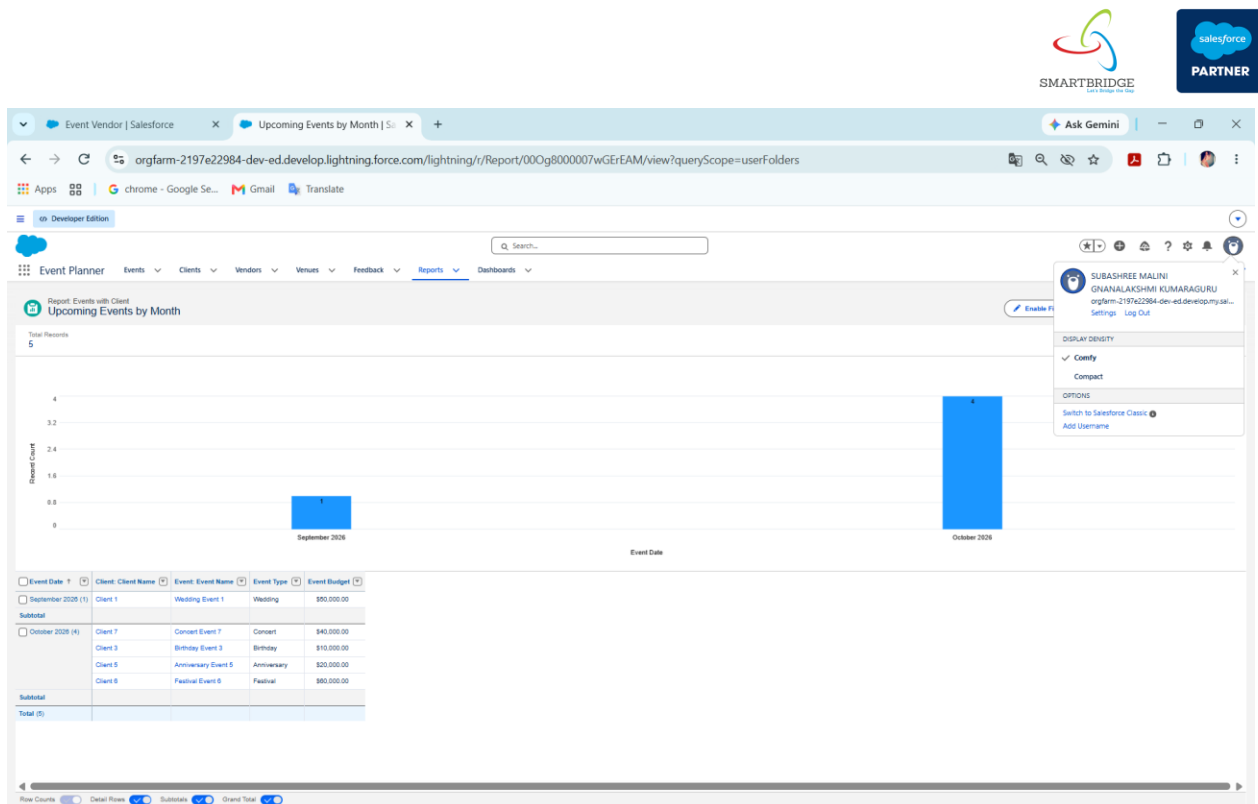
The **Event Planner** Lightning App was created to provide a centralized interface for managing the EventForce Management System. The app includes navigation items for Events, Clients, Vendors, Venues, Feedback, Reports, and Dashboards. This allows users to access the main event management functions from a single application.



6.2 Reports & Dashboards

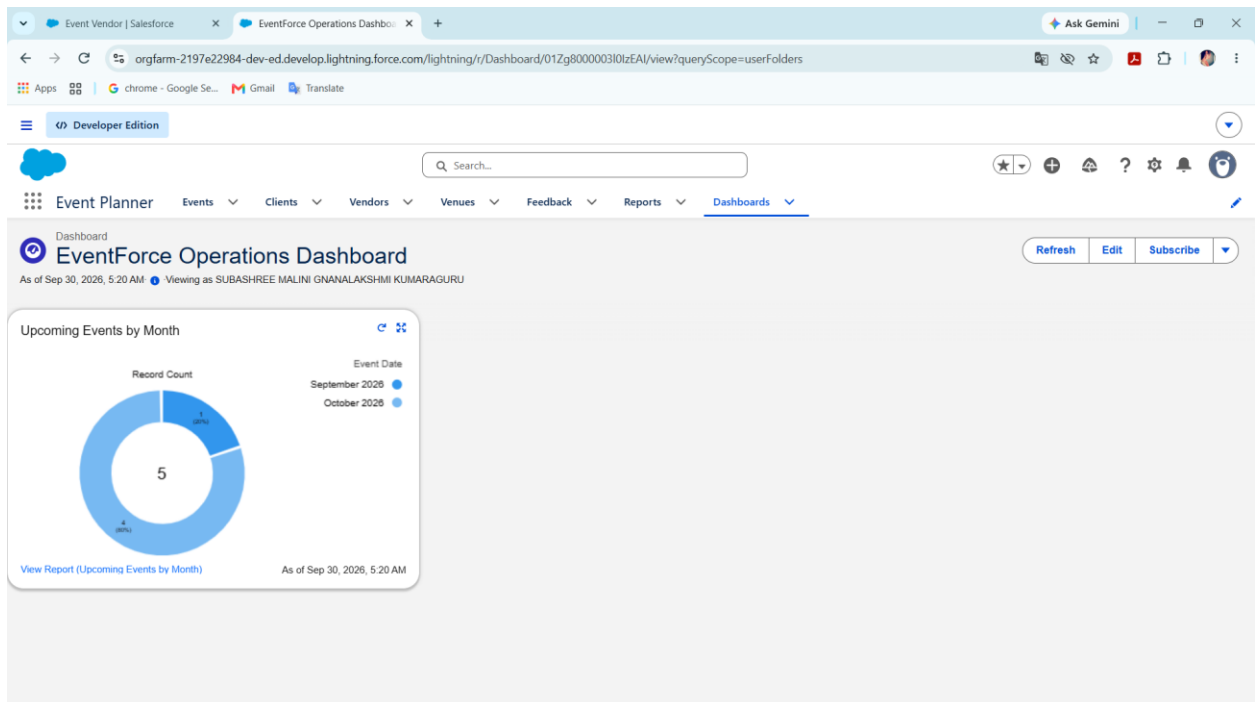
Report:

The **Upcoming Events by Month** report was created to provide visibility into upcoming events. The report groups events by the calendar month of the Event Date and displays the number of records along with the total Event Budget. The report uses upcoming event criteria and excludes cancelled events.



Dashboard:

The **EventForce Operations Dashboard** was created to provide a visual summary of upcoming events. It uses the Upcoming Events by Month report as its source and presents the information using a donut chart for quick operational analysis.



7. Phase 4 – Data Migration, Testing & Security

7.1 Data Migration

Data Migration was performed to load validated records into Salesforce for Events, Clients, Vendors, Venues, and Feedback. Data was validated before import to reduce duplicate records and maintain consistency. The Salesforce Data Import Wizard was used for importing the required records.

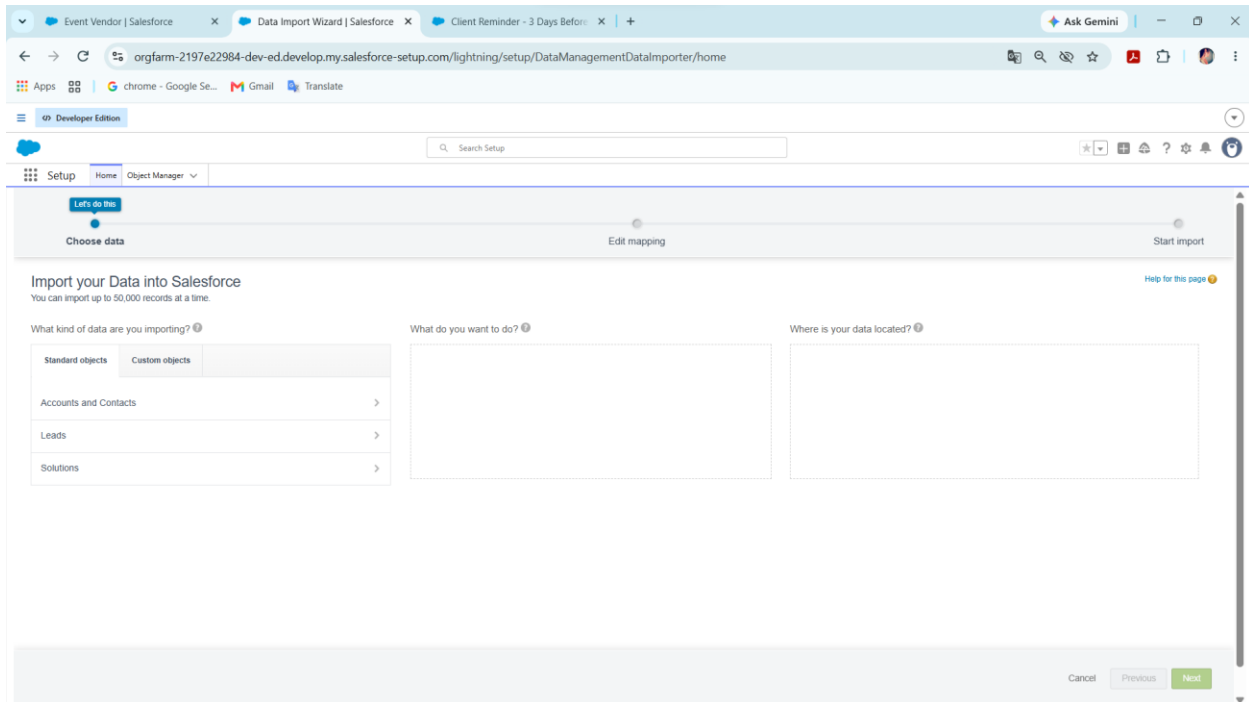


Figure 7.1 – Salesforce Data Import Wizard

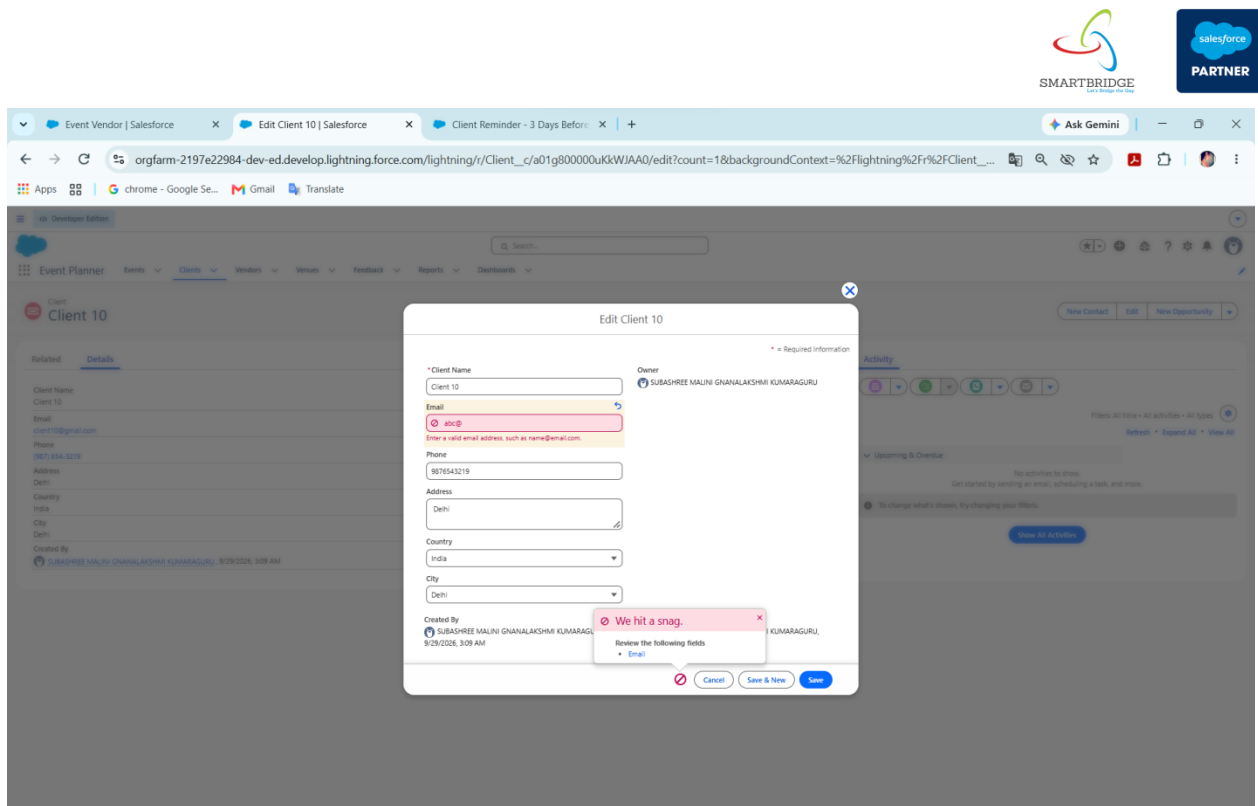


Figure 7.2 – Client Email Validation Testing

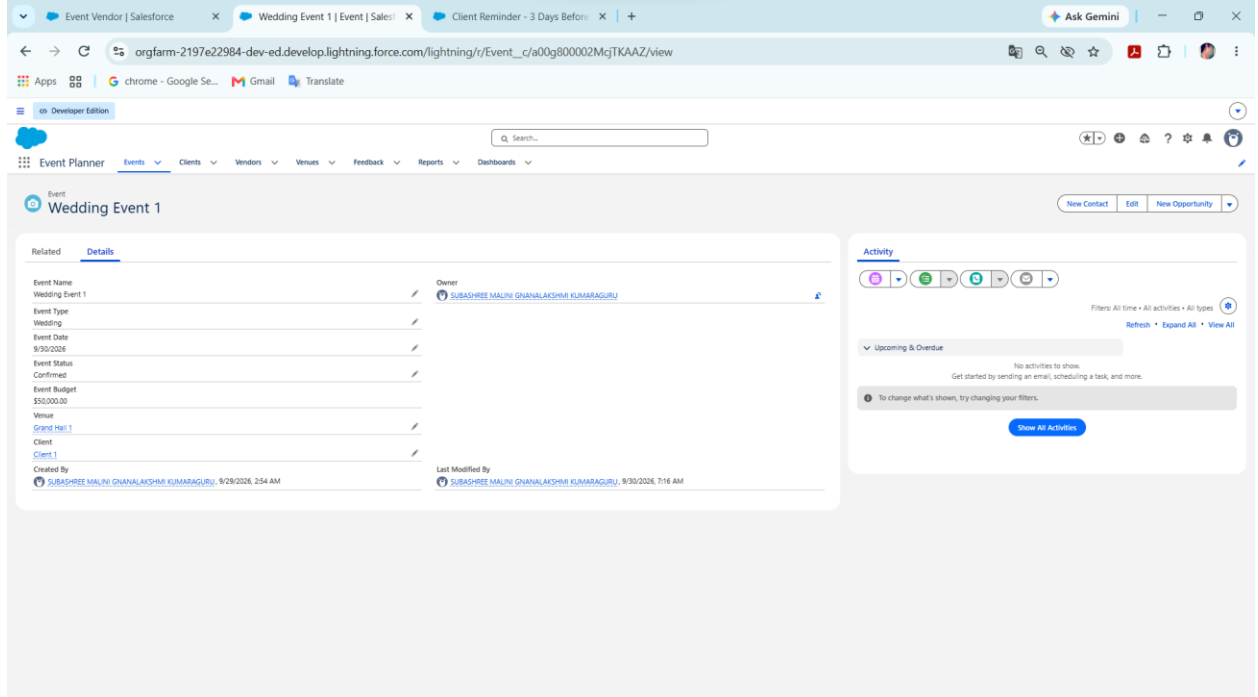


Figure 7.3 – Confirmed Event Automation Testing

7.2 Functional & Performance Testing

Security Testing:

Security configurations were verified using Profiles, Roles, Permission Sets, Organization-Wide Defaults, and Sharing Rules. These configurations were reviewed to ensure that users receive access according to their assigned responsibilities.

8. PHASE 5 – DEPLOYMENT, MAINTENANCE & DOCUMENTATION

8.1 Deployment Strategy

The EventForce Management System was developed and tested in a Salesforce Developer Edition environment. Production deployment was not performed as part of this academic implementation.

For a real-world implementation, the configured Salesforce components can be moved from the development environment to a production Salesforce environment using appropriate deployment methods. Deployment may involve Sandboxes, Change Sets, Salesforce DX, or DevOps/CI-CD tools.

The deployment strategy would include:

- Moving custom objects, fields, relationships, and validation rules.
- Deploying Flows, Approval Processes, Apex Classes, and Triggers.
- Deploying Profiles, Permission Sets, Roles, and Sharing Rules.
- Deploying Reports, Dashboards, and Lightning App configurations.
- Performing testing in the target environment before production release.

8.2 Maintenance Approach

Regular maintenance is important to keep the EventForce system reliable as business requirements change.

The maintenance activities would include:

- Monitoring and troubleshooting Flows and scheduled automation.
- Reviewing Apex Classes, Triggers, Batch Apex, and Scheduled Apex.
- Updating Validation Rules and Approval Processes when requirements change.
- Reviewing Profiles, Permission Sets, Roles, OWD, and Sharing Rules.
- Maintaining Reports and Dashboards based on changing operational requirements.
- Monitoring data quality and correcting inconsistent records.
- Reviewing system performance and automation behaviour periodically.

The source document specifically identifies ongoing maintenance and monitoring of Flows, Apex jobs, Approval Processes, Validation Rules, and Sharing Settings as important maintenance activities.

8.3 Testing and Post-Deployment Verification

After deployment, the EventForce configuration should be verified to ensure that the major business processes continue to work correctly.

The verification areas include:

- Client email validation.
- Confirmed event reminder automation.
- Event cancellation approval process.
- Venue double-booking prevention.
- Venue availability updates.
- Batch and scheduled event completion.
- User access and record visibility.
- Reports and Dashboard functionality.

8.4 Documentation

The project documentation provides a consolidated record of the EventForce implementation. It includes:

- Project Overview and Objectives.
- Requirement Analysis and Problem Statements.
- Custom Objects, Fields, and Relationships.
- Validation Rules and Formula Fields.
- Flows and Approval Processes.
- Apex Classes, Triggers, Batch Apex, and Scheduled Apex.
- Lightning App, Reports, and Dashboard.
- Data Migration, Testing, and Security configuration.
- Project Planning and Scheduling.
- Conclusion, Future Scope, and Appendix.
- Screenshots supporting the major Salesforce configurations.

8.5 Future Deployment and Enhancement

The current EventForce implementation provides a foundation for future deployment and enhancement. Future activities can include production deployment, advanced event automation, enhanced vendor management, advanced analytics, external integrations, mobile accessibility, scalability improvements, and continuous system optimization.

9. Project Planning

The EventForce Management System project was planned using an Agile-based approach with sprint-wise execution. The project requirements were organized into functional areas and user stories, with story points used to estimate the effort required for each task. Sprint-based planning was used to organize the development activities related to Event Management, Client and Venue Management, Vendor Management, Notifications and Automation, Approval Management, Security and Access Control, and Reports and Dashboards.

The project planning approach helped organize the implementation activities into manageable development tasks and provided a structured way to track progress throughout the project.

9.1 Sprint Planning / Schedule

Sprint	Total Story Points	Duration	Sprint Start	Planned End	Story Points Completed	Actual Release
Sprint-1	20	8 Days	11 Apr 2021	18 Apr 2021	20	19 Apr 2021
Sprint-2	20	6 Days	12 May 2023	17 May 2023	—	—
Sprint-3	20	5 Days	14 Nov 2023	18 Nov 2023	—	—
Sprint-4	20	4 Days	11 Dec 2023	14 Dec 2023	—	—

9.2 Project Tracker, Velocity & Burndown Chart

The project progress was planned and tracked using sprint-wise story points. The project tracker records the planned story points, sprint duration, start date, planned end date, completed story points, and actual release date. Velocity and burndown tracking can be used to monitor sprint progress and support future estimation.

10. Advantages & Disadvantages

10.1 Advantages

- **Centralized Data Management** – Event, Client, Vendor, Venue, Feedback, and EventVendor information is managed within Salesforce.
- **Process Automation** – Flows automate reminders and other business processes, reducing manual effort.
- **Prevents Double Booking** – Apex Trigger and VenueStatusHelper help prevent venue booking conflicts.
- **Controlled Cancellation Process** – Approval Process provides structured handling of event cancellation requests.
- **Improved Operational Visibility** – Reports and Dashboards provide visibility into event schedules, vendor activities, and feedback.
- **Secure Access** – Profiles, Roles, Permission Sets, OWD, and Sharing Rules provide controlled access to records.
- **Scalable Solution** – Salesforce can support future growth and additional business requirements.
- **Better Data Quality** – Validation Rules and data validation help maintain accurate and consistent data.

10.2 Disadvantages

- **Developer Edition Limitation** – The project was implemented in Salesforce Developer Edition for academic purposes and was not connected to a production environment.
- **Production Deployment Not Performed** – Actual migration to production was outside the scope of this implementation.
- **Deployment Requires Additional Tools** – Real-world deployment may require sandboxes, Change Sets, Salesforce DX, or DevOps/CI-CD tools.
- **Configuration Complexity** – Managing Flows, Apex, Approval Processes, and security configurations requires appropriate Salesforce knowledge.
- **Ongoing Maintenance Required** – Flows, Apex jobs, approval processes, validation rules, and sharing settings need regular monitoring and troubleshooting.

11. Conclusion

The EventForce Management System demonstrates how Salesforce can be used to streamline and automate event planning and management operations. The project brings Events, Clients, Vendors, Venues, Feedback, and EventVendor data into a centralized platform.

The implementation combines Salesforce Lightning, Flows, Validation Rules, Formula Fields, Approval Processes, Apex Triggers, Apex Classes, Batch Apex, Scheduled Apex, security controls, Reports, and Dashboards to address key business requirements. Automation supports activities such as event reminders, cancellation approvals, venue availability management, double-booking prevention, and event completion.

The project also demonstrates the importance of data migration, testing, security, documentation, and maintenance in a Salesforce implementation. The solution was developed and tested in a Developer Edition Org, and production deployment was not performed as part of the implementation.

12. Future Scope

The EventForce Management System can be further enhanced to support larger-scale event operations and improve automation, integration, and user experience.

- **Production Deployment** – Deploy the solution from the development environment to a production Salesforce environment using appropriate deployment tools.
- **Advanced Event Automation** – Extend automation for additional event planning and operational activities.
- **Enhanced Vendor Management** – Introduce more advanced vendor tracking and performance management capabilities.
- **Advanced Analytics** – Add more Reports and Dashboards for deeper insights into events, venues, vendors, and client feedback.
- **External Integrations** – Integrate Salesforce with external systems and services required for real-world event operations.
- **Mobile Accessibility** – Enhance the solution for convenient event management through mobile devices.
- **Scalability Enhancements** – Expand the system to support larger numbers of events, clients, vendors, venues, and users.
- **Continuous Maintenance & Optimization** – Regularly improve Flows, Apex, validation rules, sharing settings, and other configurations as business requirements evolve.

13. APPENDIX

A. Custom Objects

- Event
- Client
- Vendor

- Venue
- EventVendor
- Feedback

B. Automation

- Record-Triggered Flow – Client Reminder - 3 Days Before
- Validation Rule – Email_Valid_Address
- Formula Field – Event Budget
- Approval Process – Event_Cancellation_Process
- Apex Trigger – EventTrigger13
- Apex Trigger – PreventDoubleBooking
- Apex Class – VenueStatusHelper

C. Processing

- Batch Apex – BatchCompleteEvents
- Scheduled Apex – ScheduleCompleteEvents
- Scheduled Job – Daily Event Completion

D. Security

- Profiles – Event Admin, Event Coordinator, Vendor Manager, Client
- Roles – Event Admin, Event Coordinator, Vendor Manager, Client
- Permission Set – Feedback Manager
- Organization-Wide Defaults – Event Private
- Sharing Rule – Event_Sharing_For_Vendors

E. Data Migration

- Salesforce Data Import Wizard

F. UI & Analytics

- Lightning App – Event Planner
- Report – Upcoming Events by Month
- Dashboard – EventForce Operations Dashboard

G. Testing & Documentation

- Client Email Validation Testing
- Confirmed Event Flow Testing
- Venue Double-Booking Testing
- Security Configuration Testing
- Project Documentation

Thank You!!